The Pension and the Drawdown Rule

Why the All-Equity Lifecycle Portfolio Is Conditional on Both

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Abstract

A lifecycle investor holding only equities, split between domestic and international markets, is said to dominate the age-declining glide path embedded in target-date funds. We show that this prescription is conditional on two institutions rather than on the return process: the public pension the investor retires onto, and the rule they draw the portfolio down by. Neither settles it alone. On a 16-country panel of real returns spanning 1890–2020, simulated with a calendar-joint block bootstrap, we cross three pension systems with 8 withdrawal rules and score every portfolio in each cell on the same simulated lifetimes.

The all-equity portfolio leads the target-date fund by 11.09% in certainty-equivalent retirement consumption under the United States' earnings-related schedule. Replacing that schedule with Australia's — a means-tested Age Pension alongside a compulsory 12% Superannuation Guarantee — reverses the ordering to -2.08%, and the target-date fund wins. That reversal is the paper's starting point and not its conclusion, because it is one cell of 24. It appears under 1 of 8 withdrawal rules, and that rule is the fixed real withdrawal the literature spends by — which on the same household delivers less than half the certainty-equivalent consumption of the best rule in the same menu, at 12 percentage points more ruin. Under every other rule the all-equity portfolio leads in Australia too, by up to 27%.

A delete-one-country jackknife over the sixteen markets settles which of those two facts the panel can actually support. The reversal carries a standard error of 10.3 points and an interval of [-22.4, +18.2] that contains zero, and its sign changes in 8 of the sixteen sub-panels. The rule effect does not: every other cell has a standard error between 2 and 5 points and an interval excluding zero. Paired across the same sixteen deletions, the largest rule effect is +28.7 points with an interval of [+8.8, +48.6], and 6 of 7 such differences exclude zero. So the finding this cross-section supports is the interaction, not the reversal — which we report as a point estimate a sixteen-country panel is too small to separate from no difference at all.

Re-scoring the same simulated lifetimes at risk aversions of 2, 5 and 10 adds a third condition, and it is the one we did not expect. The contested cell runs +21.91% at $\gamma = 2$, -2.08% at $\gamma = 5$ and -36.40% at $\gamma = 10$: the reversal is absent at the lowest risk aversion scored and turns over between there and the baseline. Every other rule in the menu is positive at every risk aversion. The all-equity prescription therefore fails only where a means-tested pension, a fixed real withdrawal and a sufficiently risk-averse household meet, which is a narrower claim than the one we set out to make and a more useful one: two of those three are institutions a plan sponsor chooses, and the third is not.

Every gap quoted here compares two *portfolios* facing the same simulated lifetimes; where we instead compare two countries' consumption levels, or two all-equity portfolios against each other, we say which.

The mechanism is not the means test's taper. This household holds 32.1 times average earnings against a cut-off of 6.8, and 98% of simulated households are past the test before it is applied. What the means test removes is an unconditional real annuity, and with it the floor under bad outcomes: mean retirement consumption *rises* +27% while the fifth percentile *falls* -27%. Paying the same flat pension unconditionally, means-testing nothing, leaves the all-equity portfolio ahead by +12.78% — the ordering is intact even though that pension pays this household less than the American schedule does. It is the test that reorders portfolios, not the pension's level.

A floor need not come from a pension, and the drawdown rule can supply one. Holding the pension and the returns fixed and varying only the withdrawal rule, the all-equity lead in the Australian system moves from -2.08% under the fixed real rule the literature spends by to +26.63% under amortisation at 8%, which cannot deplete the portfolio and so manufactures the floor the pension no longer provides. The portfolio ordering therefore reverses a second time. The all-equity prescription is conditional on two institutions at once, and is silent about a household that has neither a guaranteed income nor a rule that cannot run out.

Two further results are needed before that reading can be trusted. Charging the Superannuation Guarantee to wages rather than to the employer costs 13.4% of lifetime certainty-equivalent consumption and leaves the balance at retirement, and therefore the retiree's whole problem, exactly unchanged — so the cross-system comparisons are bracketed by that figure rather than pinned. And scaling the balance until the assets test does bind, the retiree wants 0% equity at every position under a fixed real withdrawal and 100% under a percentage-of-balance rule — a result that survives risk aversions from 2 to 10 and consumption floors three orders of magnitude apart. The portfolio a means-tested retiree should hold is not a property of the means test; it is a property of the means test and the withdrawal rule jointly.

Keywords: lifecycle asset allocation; public pension design; means testing; target-date funds; certainty-equivalent consumption; decumulation. **JEL:** G11, G51, H55, D14.

1. Introduction

The default investment vehicle of the modern retirement system is the target-date fund, and it embodies one proposition: that an investor should hold less equity as they age. Anarkulova, Cederburg and O'Doherty (2023) challenge it directly, showing that when returns are drawn in blocks from the international historical record rather than i.i.d. from post-war US data, a fixed all-equity portfolio beats the glide path on almost every metric an investor would care about. The finding has been influential precisely because it is robust to the things a reader first reaches for: it survives higher risk aversion, a longer horizon, and a return panel that includes every disaster the twentieth century supplied.

It is not robust to the thing nobody reaches for, which is the pension. That result is derived for an investor who receives the United States' Social Security: an earnings-related benefit, paid whatever else the retiree owns. Roughly a third of OECD members instead pay a residence-based benefit that is withdrawn against the retiree's own assets or income, Australia's Age Pension among them. This paper asks what the all-equity prescription becomes for an investor under that second kind of scheme, and finds that it reverses.

The exercise is deliberately narrow. Nothing about the return process changes between our two arms: the same panel, the same bootstrap, the same simulated lifetimes, the same household, the same fund menu. Only the benefit formula moves. Whatever difference appears is therefore attributable to the pension and to nothing else, which is what lets the paper say *why* the prescription reverses rather than only *that* it does.

1.1 What we find

The ordering reverses under an asset-tested pension. The all-equity portfolio's lead over the target-date fund, +11.09% under the American schedule, becomes -2.08% when that schedule is replaced by Australia's, and the de-risking glide path takes first place. Nothing about the return panel changes between those two runs; the objective function does. Two qualifications belong with that sentence rather than after it. The reversal holds under 1 of the 8 withdrawal rules we run; under every other one the all-equity portfolio leads in Australia too. And the reversal is the one cell of the table this cross-section cannot resolve: a delete-one-country jackknife over the sixteen markets puts a standard error of 10.3 points on a point estimate of -2.08%, and the sign changes when eight of the sixteen are removed one at a time. Every other cell in that table has a standard error between 2 and 5 points and an interval excluding zero.

A third condition is the risk aversion. Re-scoring the same simulated lifetimes at risk aversions of 2, 5 and 10, the contested cell runs +21.91% at $\gamma = 2$, -2.08% at $\gamma = 5$ and -36.40% at $\gamma = 10$: the reversal is absent at the lowest of the three and deepens through the other two, while every other withdrawal rule stays positive at all three. So the failure of the all-equity prescription is located precisely — an asset-tested pension, a fixed real withdrawal, and a household risk-averse enough to pay for the floor — and two of those three are chosen by whoever sets the defaults.

The mechanism is the floor, not the taper. A means test is usually discussed as an implicit tax on wealth, and Australia's is steep enough to qualify — a dollar of assessable assets costs 7.8% of pension a year, more than domestic equity earns on average in this panel. But that reading does not survive checking where the household stands: it is past the cut-off before the test is applied, so the taper reaches it only in the left tail, after a portfolio has already fallen. What the means test removes is the unconditional annuity itself. A retiree standing on a guaranteed floor can carry the equity tail; one standing on their portfolio alone cannot, and a risk-averse objective prices that difference heavily.

A floor need not come from a pension. The reversal is conditional on a withdrawal rule that can run out. Under an amortisation rule the balance is divided by the years remaining, ruin falls to zero, and the household manufactures its own floor out of the larger portfolio compulsory saving bought it. Holding the pension and the returns fixed and changing only the rule, the all-equity lead in the Australian system moves from -2.08% to +26.63% under amortisation at 8%, so the portfolio ordering reverses a second time. The all-equity prescription is conditional on two institutions rather than one.

It is worth being exact about what that sentence compares, because two orderings are easy to run together. One is between *portfolios*: which of two funds a given retiree should hold. The other is between *countries*: which pension system delivers more retirement consumption. They can move in opposite directions, and the claim above is about the first. Every percentage in this paper is a portfolio comparison unless it is labelled otherwise.

Who pays for compulsory saving changes the level and not the portfolio. Australia's Superannuation Guarantee is levied on the employer, and modelling it that way hands the Australian arm income the American arm never receives. Charging it to wages instead costs 13.4% of lifetime certainty-equivalent consumption — so every comparison here is bracketed by that figure — but it leaves the balance at the pension age, and therefore the entire retirement problem, exactly unchanged. The contribution is made either way. That is an identity rather than an estimate, and it means the incidence question and the portfolio question are separate questions.

On a household the test actually binds, the portfolio is a property of the withdrawal rule, not of the pension. The household simulated here retires on 32.1 times average earnings, so we scale the arriving balance until it sits inside the taper band and ask again. Spending a fixed real amount, it wants no equity at any balance the test can reach; spending a share of the current balance, it wants 100% equity at every one of them. The reason is mechanical and, once seen, obvious: under a fixed real rule a good return is never spent, so it accumulates into assessable assets, the pension is withdrawn against them at 7.8% a year, and consumption does not rise at all. The upside is confiscated and the downside is not.

1.2 Contribution

Three things here are new, and it is worth separating them from what is replication. That an all-equity portfolio beats a glide path on an international block bootstrap is Anarkulova, Cederburg and O'Doherty's result, and Section 5 reproduces it rather than claiming it.

First, the *conditionality*: the prescription is a property of the pension the investor retires onto, and we can say which feature of the pension by paying the same benefit unconditionally, which leaves the all-equity portfolio ahead by +12.78%. Withdrawing a pension at the margin and paying a smaller one are different interventions, and only the first reorders portfolios. Second, the *mechanism*: Section 2 writes the retiree's budget line as three regimes and shows that inside the taper band a means test is insurance rather than a wealth tax — a prediction sharp enough to fail, which under one of the two withdrawal rules it does. Third, the *interaction*: the portfolio and the drawdown rule are one decision when the pension is asset-tested, and we measure how much of one.

1.3 Relation to the literature

The lifecycle portfolio-choice literature since Cocco, Gomes and Maenhout (2005) and Gomes and Michaelides (2005) treats public pension income as a bond-like endowment that crowds fixed income out of the financial portfolio. That is right for an earnings-related schedule. It is the wrong intuition for an asset-tested one, where the endowment is withdrawn precisely as the portfolio grows, and where — as Hubbard, Skinner and Zeldes (1995) show for saving — the binding feature is the floor the transfer provides rather than its expected level. Dahlquist, Setty and Vestman (2018) solve for a default fund's allocation given a pension system; we vary the system and hold the fund menu fixed, which isolates the same interaction from the other side.

The Australian assets test has its own literature, and it reaches the pension-as-endowment question from the institutional side. Iskhakov, Thorp and Bateman (2015) solve an Australian retiree's problem with the means test in the constraint set and find the taper distorts drawdown as well as portfolio choice; Bateman et al. (2018) document what Australian members actually hold. On the theory side, Sefton, van de Ven and Weale (2008) and

Braun, Kopecky and Koreshkova (2017) analyse means-tested transfers as insurance rather than as taxes, and Butler, Peijnenburg and Staubli (2013) show a means-tested benefit crowding out voluntary annuitisation for exactly the reason the floor matters here. What we add to that work is the comparative statement — the same household, the same returns, two pension systems — and the interaction with the drawdown rule, which we have not found measured.

On the decumulation side the rule that supplies the floor is the actuarial or amortisation rule of Milevsky and Huang (2011). The withdrawal-rule literature typically conditions on a fixed income floor and the pension literature on a fixed withdrawal rule, which is why the interaction between them has room to be new.

The means-testing literature itself is largely about saving and labour supply rather than portfolio choice, and it has long understood the taper as an implicit tax. Hubbard, Skinner and Zeldes (1995) is the closest antecedent to what we find: their asset-tested transfer discourages saving not because of its expected level but because of the floor it puts under consumption. We measure the same feature acting on the composition of the portfolio rather than its size, and find that it acts in the opposite direction to the tax intuition — the taper, inside the band, makes equity cheaper rather than dearer.

Our panel is the Jordà–Schularick–Taylor macrohistory database, the same source the replicated study uses, and the same 16 developed markets Dimson, Marsh and Staunton (2002) cover for a comparable period. Every number below is regenerated from that panel by the pipeline described in Section 4; the fuller robustness apparatus, and the extensions not needed for this argument, are in the companion study.

1.4 Roadmap

Section 2 sets out the retiree's budget line under an asset test and derives the prediction the rest of the paper tests. Sections 3 and 4 describe the panel and the simulation. Section 5 reproduces the all-equity result under the American schedule, and Section 6 reverses it under the Australian one. Section 7 separates the floor from the taper; Section 8 charges the guarantee and moves the household onto the test; Section 9 finds the withdrawal rule a retiree should actually use, and Section 10 asks what that rule does to the portfolio ordering and how precisely sixteen countries can resolve each answer. Section 11 says what would change these conclusions. The two systems also differ in their tax treatment, and Section 34 of the companion study rules that out as an explanation; we do not repeat it here.

2. What an Asset Test Does to a Retiree's Budget Line

Write W for the assessable balance a household holds at the start of a retirement year, R for the gross real return it earns over that year, x for what its withdrawal rule pays out, A for the assets-test free area, b for the full pension and τ for the rate at which the pension is withdrawn against assets above A. The test is assessed on the assets that remain once the year's withdrawal has been taken, which is the convention the Australian scheme uses and the one the simulation implements. Consumption in the year is the withdrawal plus whatever pension the test leaves:

$$c = x + \min\{b, \max[0, b - \tau(WR - x - A)]\} \quad (1)$$

The three regimes are the three branches of that expression, and writing them out is what makes the mechanism visible. Let $S = WR - x$ be the assets the test sees. Below the free area ($S \leq A$) the pension is paid in full; inside the band ($A < S < A + b/\tau$) it is withdrawn at the taper; above the cut-off it is gone:

$$c = x + b \quad \textit{(below the free area)} \quad (2)$$

$$c = x(1 + \tau) - \tau WR + (b + \tau A) \quad \textit{(inside the taper band)} \quad (3)$$

$$c = x \quad \textit{(above the cut-off)} \quad (4)$$

2.1 Why the withdrawal rule decides the sign

A means test is usually described as an implicit tax on wealth, and Australia's is steep enough to qualify: at 7.8% a year it exceeds what domestic equity earns on average in this panel. The middle line says that description is incomplete, and what it puts in its place depends entirely on how x responds to R. Differentiate it with respect to the return:

$$\partial c/\partial R = (1 + \tau)\,\partial x/\partial R - \tau W \quad (5)$$

Take the rule that spends a fixed real amount first, because it is the one the literature assumes and the one this paper spends by. Here x is a constant, so $\partial x/\partial R = 0$ and the derivative is $-\tau W$, negative at every withdrawal rate. A good return raises assessable assets and leaves the withdrawal exactly where it was — this year and every year after it, because the rule does not read the balance at all. The gain is taxed at 7.8% a year for as long as it is held and reaches consumption in no year at all; the only place it can surface is the estate. The upside is confiscated and the downside is not.

Now the rule that spends a fixed share k of the current balance. Here $x = kWR$, so $\partial x/\partial R = kW$ and the derivative is $W[k(1 + \tau) - \tau]$. This is positive only above a threshold withdrawal rate, $k > \tau/(1 + \tau)$, which on the Australian taper is 7.24%. It is worth being blunt that the proportional rule this paper actually simulates draws 4% and therefore sits *below* that threshold: in the year a good return arrives, this household too hands the test more in withdrawn pension than it takes in extra spending.

The two rules nonetheless differ in the way that matters, and the threshold is what makes the difference visible rather than what creates it. Under the proportional rule the gain is deferred: the larger balance is still there, and every later withdrawal is a share of it, so the return reaches consumption eventually whatever k is. Under the fixed real rule the deferral never ends, because no future withdrawal reads the balance either. A proportional rule spends the gain late; a fixed real rule never spends it. That is the distinction which survives from one period to a retirement, and the threshold only says how late.

2.1.1 What that implies for the portfolio

The three regimes then order the retiree's problem, and the comparison is two lines. Exposure to the return is $\partial x/\partial R$ below the free area, $(1 + \tau)\partial x/\partial R - \tau W$ inside the band, and $\partial x/\partial R$ again above the cut-off. The guaranteed part of consumption goes b, then $b + \tau A$, then nothing. So inside the band the household faces strictly less exposure than below the free area and stands on a strictly higher floor; above the cut-off it faces the same exposure as below the free area and stands on no floor at all.

For an investor with constant relative risk aversion, a larger riskless component of consumption raises the optimal share of the risky asset — the standard result for a bond-like endowment, and the one the lifecycle literature invokes to explain why a pension crowds fixed income out of the financial portfolio. Applying it to the ordering above gives the prediction directly: **optimal equity should be highest inside the taper band, lower below the free area, and lowest above the cut-off** — non-monotone in wealth, with its minimum where the pension has been withdrawn entirely rather than where it is being withdrawn.

That is the prediction under a rule that spends the balance. Under a fixed real rule the insurance term never arrives, because the household never converts the return into anything it eats, and the prediction is a corner rather than a shape: no equity anywhere the taper operates, and position against the test should barely matter, because what does the damage is the rule and not the band.

On this calibration the free area is 3.01 times average earnings, the full rate 29.3% of them, and the cut-off 6.77. Section 8 runs both rules across all three regimes and reports which prediction holds.

2.2 What the model does not settle

This is one period of accounting, not a solved lifecycle problem. It says which way the forces point; it does not say how large the non-monotonicity is where it exists, because that depends on the return distribution, the horizon and the risk aversion. Nor does one period describe the region a real means test spends most of its time in, where a household inside the band this year is over the cut-off next year and the test is re-assessed annually against a drawn-down balance. Both are simulation questions, and the simulation is the rest of the paper.

The dependence on risk aversion turns out to be the sharper of those two omissions, and worth flagging here rather than letting it arrive as a surprise. One period of accounting says the curvature of the objective scales the effect; Section 10.4 finds that over the range of risk aversions commonly used it changes the effect's *sign*. Nothing in the algebra above rules that out — the floor and the exposure move in opposite directions across the band, and which dominates is exactly the question a felicity function answers — but the algebra does not predict it either, and we did not expect it.

One caution belongs with the second prediction rather than with the results. It is a corner at zero, and a constant-relative-risk-aversion objective with a consumption floor near zero is unbounded below — a handful of near-starvation years can move a certainty equivalent further than a decade of ordinary ones. A prediction of “no equity” is exactly the kind one should distrust, so Section 8 re-scores it at several risk aversions and several consumption floors before reporting it.

3. Data

Everything below rests on one panel, and the argument of this paper is a comparison of two objective functions over it rather than a claim about the returns themselves. So this section is kept to what a reader needs to judge that comparison: what the panel is, which markets are in it, and what is known to be wrong with it. The construction detail, the source audit and the coverage arithmetic are in the companion study.

3.1 What the panel is

The panel is an annual, country-by-year matrix of five real series — domestic equity, international equity, long-term government bonds, short-term bills and consumer price inflation — for 16 developed markets over 1890–2020. It contains 2,010 usable country-years of domestic equity returns, with a median country history of 125 years and a minimum of 115 years.

All series are *real*: nominal total returns are deflated by each country's own consumer price index in the same year, so that a hyperinflation shows up as a catastrophic real return rather than as a spectacular nominal one. This is not a cosmetic choice. Several of the worst episodes in the sample — Germany in the early 1920s, central Europe in the late 1940s — are invisible in nominal space and dominate the left tail in real space.

The equity series across the panel average 6.8% in arithmetic real terms and 4.3% geometrically, with a mean cross-country standard deviation of 22.2%. The gap between the arithmetic and geometric means — over three percentage points — is itself the story: at this level of volatility, the compounded experience of a lifetime investor is far below the average annual return, and any simulation that draws i.i.d. from the arithmetic mean will overstate what a real investor would have received.

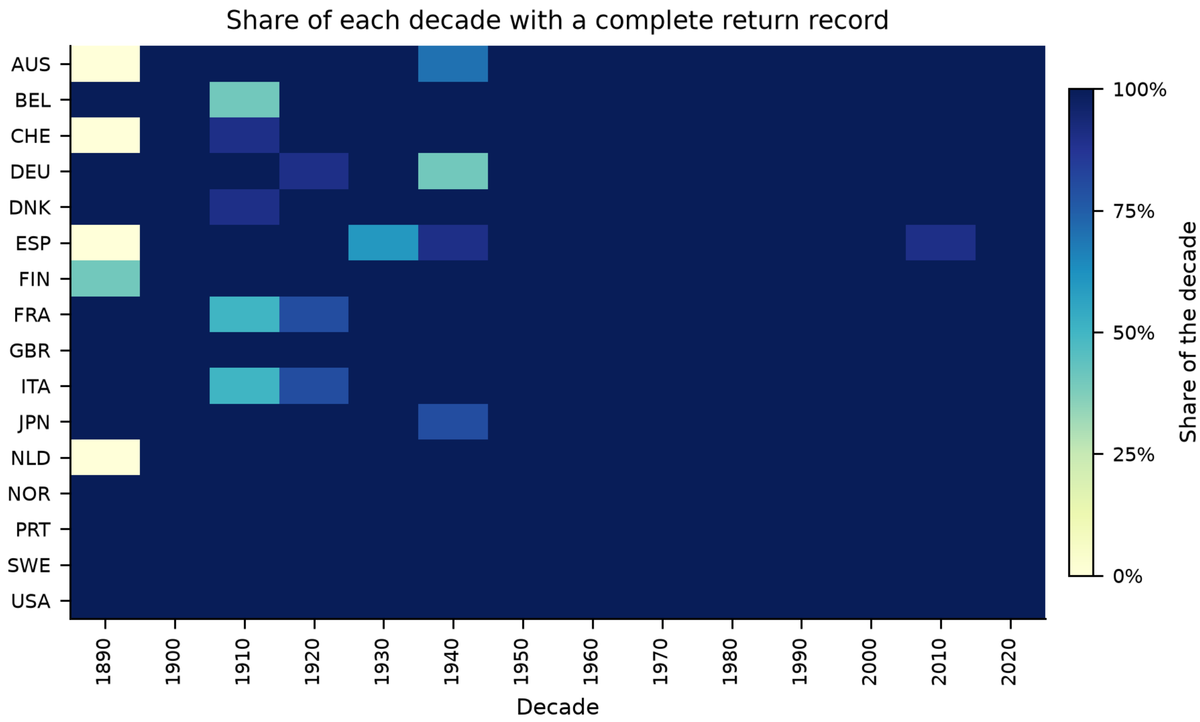

Figure 1.

Share of each decade for which a country has a complete return record. Darker is more complete; the scale is a share so that the final, partial decade compares with the rest. The gaps are not random: they cluster around the two world wars and around the market closures that follow them, which is precisely the period a survivorship-prone sample would drop.

3.2 The country set

The Jordà–Schularick–Taylor database carries complete, independently sourced equity, bond and bill total returns for 16 countries: Australia, Belgium, Denmark, Finland, France, Germany, Italy, Japan, Netherlands, Norway, Portugal, Spain, Sweden, Switzerland, United Kingdom, United States. That set is the panel, and every country-year of every return series in it is a recorded observation.

The wider developed-market universe — IMF advanced economies with an investable domestic equity market, plus Poland — numbers 38. The histories of the other 22 exist in commercial databases such as Global Financial Data and Dimson–Marsh–Staunton, which are proprietary and not redistributable. Working from openly licensed sources, 16 is the extent of the recorded evidence, and Section 3.6.1 of the companion study reports what the excluded countries do carry.

That breadth is the paper's principal limitation and Section 11.1 develops it. Its most direct consequence is on the international leg, which is a leave-one-out average and therefore spans 15 foreign markets, all advanced economies with long histories. Every statement here about international diversification is a statement about that set.

3.3 Constructing the international leg

An investor in country *i* holding "international equity" does not hold a global index that includes their own market. For each country-year we therefore build the international return as a leave-one-out average across all other countries with data that year, computed in gross terms and in a common currency before being converted back to the home investor's real terms. Leaving the home market out is what makes the domestic and international legs genuinely distinct assets rather than two overlapping slices of the same one.

Three observations in the raw international series are extreme enough to dominate any lifetime that draws them, all of them artefacts of a market reopening after wartime closure with a collapsed price index. We winsorise the international leg at the 0.1th percentile of its own distribution; the affected observations are listed in full in Table 2 so that the reader can judge the intervention rather than take it on trust.

Table 1. Real domestic equity returns by country

Country	Years	From	To	Mean	Geo. mean	S.d.	Skew	AR(1)
Australia	118	1900	2020	7.8%	6.4%	16.8%	-0.31	-0.07
Belgium	125	1890	2020	5.9%	3.8%	20.7%	-0.00	0.08
Switzerland	120	1900	2020	6.8%	5.1%	18.9%	0.30	0.10
Germany	124	1890	2020	8.7%	4.6%	28.9%	1.08	-0.01
Denmark	130	1890	2020	7.5%	6.1%	17.2%	0.43	0.04
Spain	115	1900	2020	6.2%	4.2%	21.1%	0.77	0.32
Finland	125	1896	2020	8.9%	5.0%	30.0%	1.21	0.18
France	124	1890	2020	3.6%	1.1%	23.3%	0.86	0.15
United Kingdom	131	1890	2020	6.6%	4.9%	18.7%	0.90	-0.05
Italy	124	1890	2020	6.1%	2.6%	26.8%	0.66	-0.04
Japan	129	1890	2020	7.9%	4.2%	25.8%	0.26	0.12
Netherlands	121	1900	2020	7.1%	5.1%	21.2%	0.74	0.03
Norway	131	1890	2020	5.7%	3.7%	20.1%	0.46	0.02
Portugal	131	1890	2020	3.2%	-0.1%	26.0%	0.66	0.25
Sweden	131	1890	2020	8.1%	6.1%	20.3%	0.05	0.13
United States	131	1890	2020	8.4%	6.7%	18.7%	-0.24	0.00

Notes. Arithmetic and geometric means are annual real returns deflated by each country's own consumer price index; AR(1) is the first-order autocorrelation of the annual series. Appendix B repeats this with excess kurtosis.

3.4 Cross-asset structure

The correlation structure the bootstrap must preserve is reported below. Domestic and international equity correlate at 0.43, which is high enough that international diversification is not free and low enough that it is worth having. Bonds and bills correlate at 0.72. Inflation correlates negatively with every asset, most strongly with bills (-0.66), which is the mechanism by which cash loses in real terms over long horizons.

Table 2. Pooled cross-asset correlation matrix of annual real returns

	Dom. equity	Intl. equity	Bonds	Bills	Inflation
Domestic equity	1.000	0.429	0.315	0.234	-0.121
International equity	0.429	1.000	0.110	0.117	-0.042
Bonds	0.315	0.110	1.000	0.716	-0.469
Bills	0.234	0.117	0.716	1.000	-0.662
Inflation	-0.121	-0.042	-0.469	-0.662	1.000

Notes. Pooled across all country-years in the panel. These are the moments the block bootstrap is required to reproduce; Table 4 reports how closely it does so.

Table 3. Every winsorised observation in the international leg

Year	Country	Raw return	After winsorisation
1942	Italy	1120%	301%
1945	Japan	-86%	-70%
1946	France	308%	301%

Notes. All three are markets reopening after wartime closure against a collapsed price index. Left unwinsorised, a single one of these draws would dominate the terminal wealth of any lifetime that contained it.

3.5 Market disruptions and the survivorship question

There are 16 contiguous runs of missing data across the countries of the panel, concentrated in the two world wars. The treatment of these gaps is a substantive modelling decision rather than a data-cleaning one. Interpolating across them would smooth away exactly the episodes that motivate the whole exercise; dropping the countries would reintroduce the survivorship bias the panel exists to avoid.

We do neither. Gaps are preserved as gaps, and the bootstrap is constrained to draw only blocks that lie entirely within a contiguous run of available data for the country in question. A country with a six-year wartime hole contributes blocks on either side of it but never across it. The consequence is that the sampler is *less* able to draw from countries with disrupted histories at long block lengths, which we quantify rather than assume: the run-length admissibility statistics are reported in Section 4.1.

4. Methodology

The pipeline that turns the panel into a certainty equivalent is the replicated study's, with one change that matters here: the public pension is a parameter rather than a fixture, so the same simulated lifetimes can be scored under two systems with nothing else moving between them.

4.1 The cross-country stationary block bootstrap

A simulated lifetime is 68 years long and is assembled from contiguous blocks of history. The sampler is a stationary block bootstrap in the sense of Politis and Romano (1994): block lengths are drawn from a geometric distribution with mean 10 years, truncated to [1, 40], and blocks are laid end to end until the horizon is filled.

Two features distinguish it from a textbook block bootstrap and both matter for the result.

- **Blocks are calendar-joint.** A block is a (country, start-year, length) triple, and the same triple is applied to all five series simultaneously. Domestic equity, international equity, bonds, bills and inflation for a given simulated year therefore come from the same real country in the same real year. This is what preserves the cross-asset correlation matrix of Table 2 without any of it being imposed parametrically.
- **Blocks respect data gaps.** Admissibility is enforced through pre-computed run-lengths: for each (country, year) the sampler knows how many consecutive years of complete data follow, and only draws a block that fits inside a run. A wartime hole is never bridged.

Across the production run the sampler drew 30,848 blocks with a mean realised length of 8.82 years against a target of 10, a median of 6, a ninetieth percentile of 20 and a maximum of 40. The realised mean falls short of the target for a mechanical reason worth stating: truncation at the maximum and the gap-admissibility constraint both bite from above, so the effective block length is always slightly below the nominal one.

The country for a lifetime is drawn once (*per_lifetime*) with probability proportional to the length of that country's usable history (*history*). Both choices are varied in Appendix C: redrawing the country at every block, and weighting countries uniformly, both leave the ranking unchanged. The weighting choice has little room to bite here, because every country carries between 115 and 131 usable years — history weighting and uniform weighting assign nearly the same probabilities. The choice would matter on a panel with short histories, where a country covering only one era could otherwise speak as loudly as one covering a century and a half.

Table 4. Does the bootstrap reproduce the panel it samples from?

Series	Bootstrap mean	Panel mean	Gap (bp)	Bootstrap s.d.	Panel s.d.	s.d. ratio	Bootstrap kurtosis
Domestic equity	6.86%	6.77%	8.3	22.87%	22.50%	1.016	2.9
International equity	8.52%	8.33%	19.0	22.27%	21.69%	1.027	42.5
Bonds	2.02%	1.98%	3.9	11.51%	11.81%	0.974	6.0
Bills	0.58%	0.44%	14.0	7.37%	7.99%	0.922	24.0
Inflation	5.23%	5.97%	-74.1	18.76%	37.66%	0.498	1282.5

Notes. Means agree to within twenty basis points on every series and standard deviations to within eight percent. The inflation row has an excess kurtosis in the thousands: that is the hyperinflation episodes surviving into the simulated paths, which is the intended behaviour rather than a defect.

The correlation matrix is reproduced to a maximum absolute deviation of 0.031 across the four asset-to-asset pairs that drive portfolio behaviour. The largest deviation anywhere in the matrix is 0.068 and it sits on the inflation row, where the excess kurtosis reported above makes the sample correlation itself unstable. We report both figures rather than the flattering one.

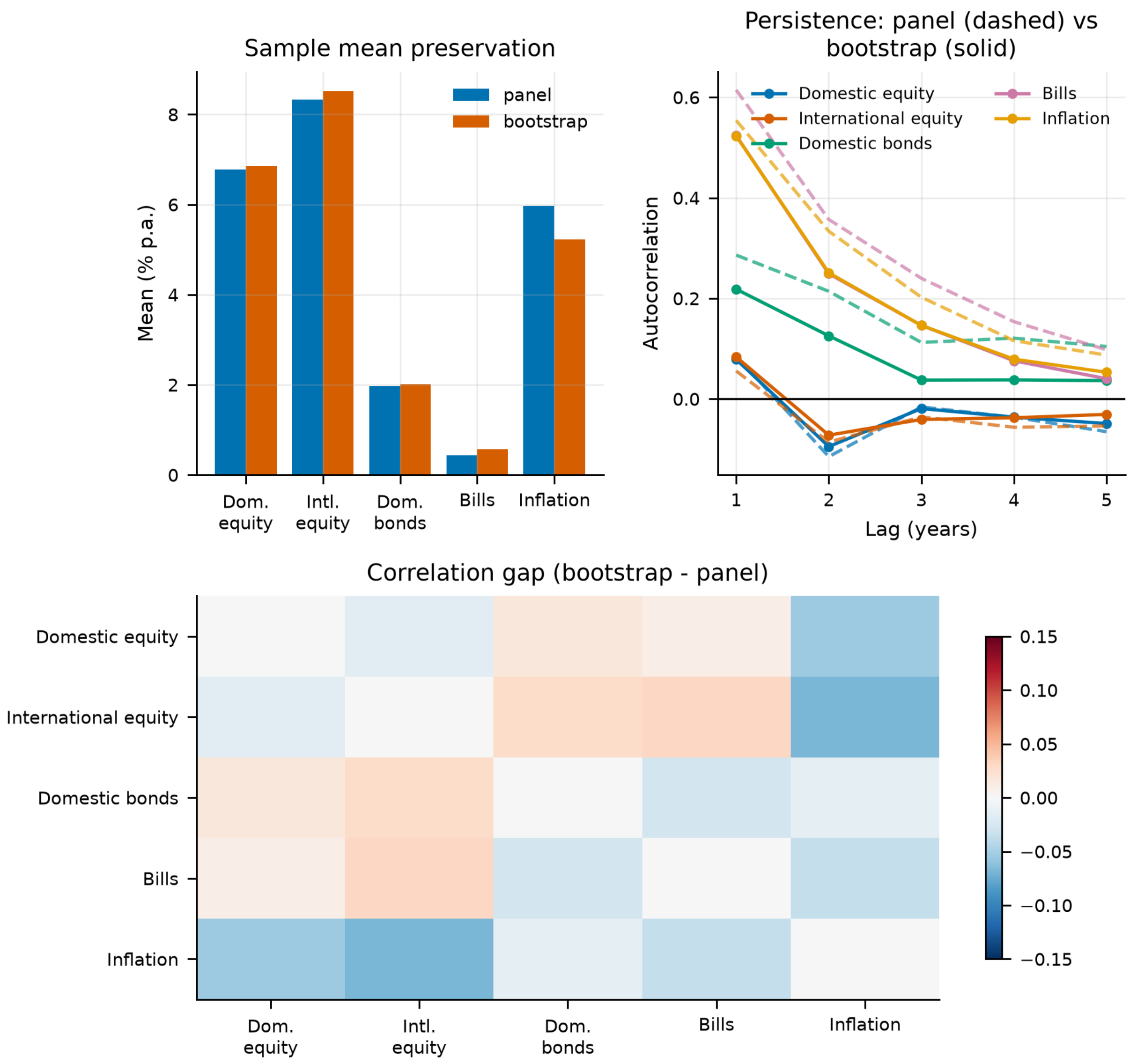

Figure 2.

Bootstrap diagnostics. Simulated marginal distributions against the panel they are drawn from, the preservation of the cross-asset correlation structure, and the within-path autocorrelation that block sampling is designed to retain.

Block length and long-horizon dispersion

Block length is the single most consequential tuning parameter in the design, because it controls how much multi-decade persistence survives into a simulated lifetime. The naive expectation is that longer blocks produce more dispersion in long-horizon outcomes. The data say the opposite, and the reason is instructive.

Table 5. Sensitivity of 68-year outcomes to the mean block length

Mean block (years)	Mean annualised	S.d. annualised	5th pct	95th pct	Within-path AR(1)
1	4.41%	3.31%	-1.52%	9.49%	-0.014
5	4.49%	3.27%	-1.41%	9.25%	0.064
10	4.43%	3.20%	-1.53%	8.89%	0.079
20	4.37%	3.14%	-1.69%	8.68%	0.088

Notes. Longer blocks raise the within-path autocorrelation, as intended, but slightly *reduce* the dispersion of annualised 68-year outcomes. Over a horizon this long the historical series mean-reverts, so preserving more of its internal structure preserves that mean reversion too.

Within-path autocorrelation rises monotonically from -0.014 at one-year blocks to 0.088 at twenty-year blocks, confirming that the sampler is doing what block sampling is for. But the standard deviation of annualised 68-year returns *falls* slightly over the same range. This is a long-horizon mean-reversion finding, not a bug: a lifetime that inherits real historical sequences inherits their tendency to revert, whereas one assembled from independent annual draws does not. It is also a caution against reading the block length as a simple dial for "more risk".

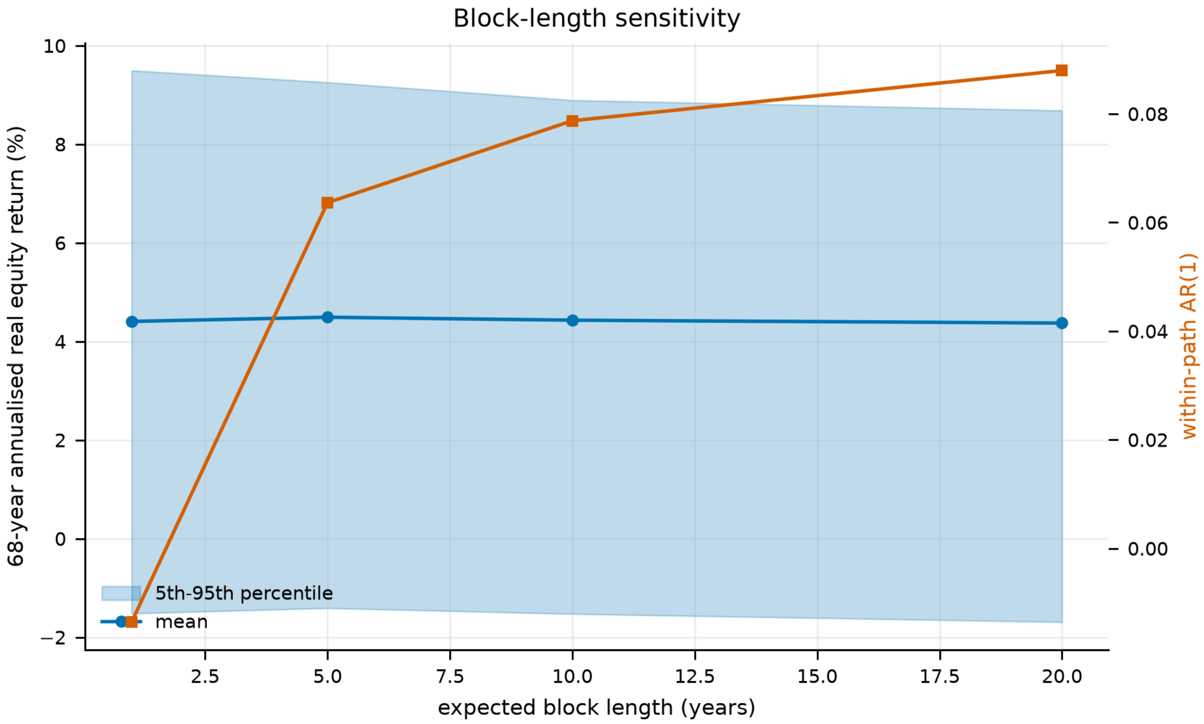

Figure 3.

Block-length sensitivity. Long-horizon dispersion is close to flat in the mean block length even as within-path persistence rises steadily, which is the signature of mean reversion in the underlying series.

4.2 The lifecycle model

A simulated investor begins work at age 25, retires at 63 and dies at 93, giving a 68-year horizon of which 38 are working years. They save a fixed fraction s = 10% of labour income, rebalance annually to the strategy's target weights, and on retirement switch to a withdrawal rule.

Wealth evolves as

$$W_{h+1} = (W_h + c_h - x_h) \cdot (1 + r^p_h), \quad r^p_h = \Sigma_a w_{a,h} r_{a,h} \quad (6)$$

where c_h is the contribution in working years and zero afterwards, x_h is the withdrawal in retirement and zero before, $w_{a,h}$ is the strategy's weight on asset a at age h, and all returns are real. Consumption is labour income net of saving while working, and the sum of the social-security benefit and the portfolio withdrawal in retirement:

$$C_h = (1 - s) Y_h \ (working) \quad C_h = B + x_h \ (retired) \quad (7)$$

Labour income follows a deterministic hump-shaped profile (b_1 = 0.045, b_2 = -0.0009 in age and age squared) multiplied by permanent and transitory shocks with standard deviations 0.10 and 0.25. The permanent shock is a cumulative sum, so income dispersion widens over a career; the transitory shock is i.i.d. The profile is an age effect and carries no economy-wide wage growth; Section 3.6.3 of the companion study measures what that leaves out and which way it biases the result.

The social-security benefit uses the US primary-insurance-amount bend-point schedule rather than a flat replacement rate, applying 90%, 32% and 15% to successive tranches of career-average earnings. This is not decoration. A flat replacement rate scales with the outcome and therefore insures nothing; a progressive schedule supplies a genuine real consumption floor, which is exactly what determines how much the left tail of a risky strategy actually costs.

Table 6. The six benchmark strategies, as constant weights

Strategy	Domestic eq.	Intl. eq.	Bonds	Bills
50/50 domestic/international equity	50%	50%	—	—
100% domestic equity	100%	—	—	—
100% international equity	—	100%	—	—
60/40 domestic equity/bonds	60%	—	40%	—
100% bills (cash)	—	—	—	100%
Target-date fund (glide path)	see below	see below	see below	see below

Notes. Held constant at every age and rebalanced annually, except the target-date fund.

The target-date fund is the one benchmark whose weights move. Its equity share is piecewise-linear in age through the knots below, interpolated between them; the equity sleeve is split 60/40 domestic/international at every age and the fixed-income sleeve 70/30 bonds/bills. The shape is the industry standard — flat and growth-heavy while young, declining through the decade before retirement, and still declining gently after it.

Table 7. The target-date fund's glide path, at its knots

Age	25	40	55	63 (retire)	75	93
Equity share	90%	90%	65%	50%	35%	30%

Notes. Linear between knots. At age 63 the fund therefore holds 30% domestic equity, 20% international, 35% bonds and 15% bills.

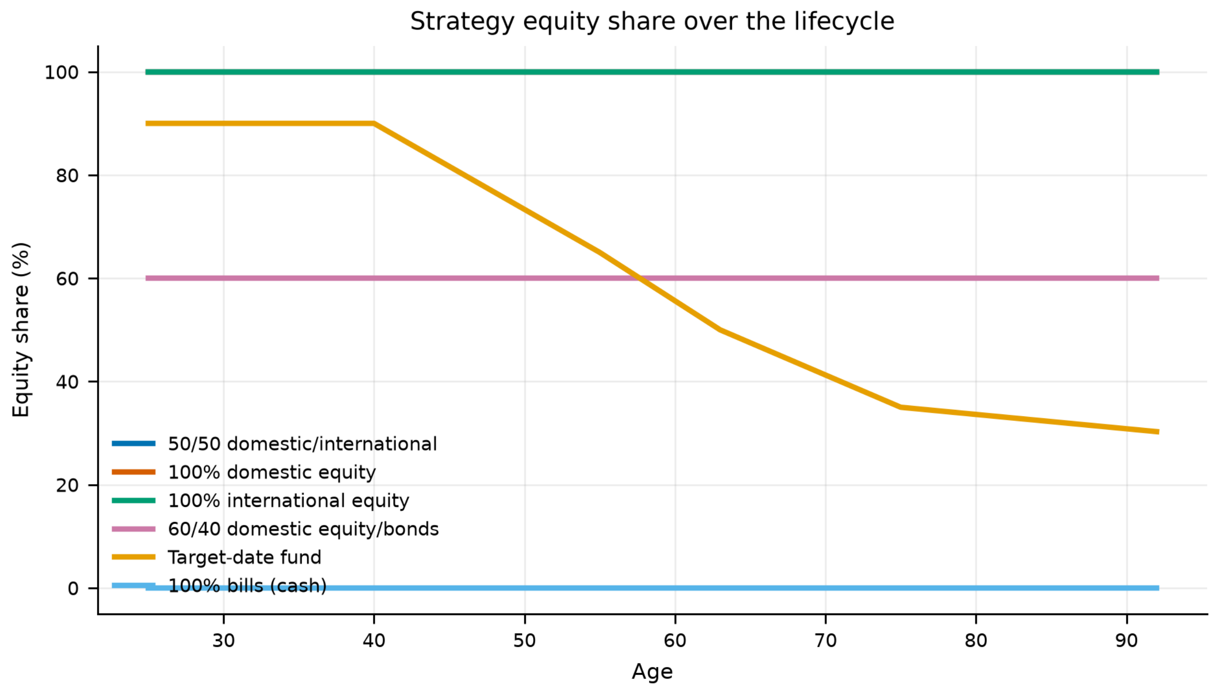

Figure 4.

The six benchmark allocation strategies as equity share by age. The target-date fund is the industry-standard declining glide path; the comparison strategies are held at fixed weights.

4.3 Preferences and the certainty equivalent

Strategies are ranked by certainty-equivalent consumption: the constant, riskless consumption stream that would leave the investor indifferent to the risky one their strategy produces. Under constant relative risk aversion with discount factor β and curvature γ, lifetime utility over the evaluation window is

$$V = \Sigma_h \beta^h u(C_h) + \beta^H \theta \cdot u(\kappa + W_H), \quad u(c) = c^{1-\gamma} / (1 - \gamma) \quad (8)$$

and the certainty equivalent inverts that utility back into consumption units. We report risk aversions of {2, 5, 10} with $\gamma = 5$ as the baseline, $\beta = 0.96$, and a bequest weight $\theta = 2$.

The bequest term uses the shifted-power specification of De Nardi (2004) with $\kappa = 1$. The shift is not cosmetic. Without it, any path that dies with exactly zero wealth contributes $u(0) = -\infty$ at $\gamma \geq 1$, one such path in a hundred thousand collapses the entire certainty equivalent, and the ranking becomes a statement about which strategy avoids exact zeros rather than about which delivers better consumption. This is a genuine trap in lifecycle work and we flag it because we walked into it.

We also report Epstein–Zin preferences in their ex-ante, early-resolution form, which separates risk aversion from the elasticity of intertemporal substitution, with ψ taking the values {0.5, 1.5}. Under CRRA these two are locked together at $\psi = 1/\gamma$, and a reader is entitled to ask whether the result is being driven by the intertemporal channel rather than the risk channel. Section 6.2 of the companion study shows it is not.

One further specification choice deserves emphasis. Utility is evaluated over the **retirement** window rather than the whole lifetime for the allocation comparisons. The reason is mechanical: with a fixed savings rate, working-life consumption is *identical* across allocation strategies by construction, so including it adds a large constant to every strategy's utility and compresses the differences that the exercise is about. Where a policy *does* change working-life consumption — the retirement timing and savings-rate studies of Sections 11 of the companion study to 28 of the companion study — we switch to the whole-lifetime window and say so explicitly.

4.4 The comparison discipline

Most of the extensions in this paper compare policies that differ in more than one way at once, and the arithmetic of a certainty equivalent will happily credit the wrong difference. Four rules are applied throughout.

- **Matched baselines.** A rule that retires people later is scored against a fixed date matched on the same *mean* retirement age, not against the original age 63. A savings rule that drifts to saving more is scored against a constant rate interpolated at its own realised career average. Without this, a policy is rewarded for working longer or saving more rather than for being smarter.
- **Common random numbers.** Every sweep and every optimiser reuses the same bootstrap draws and the same income shocks, so a difference between two settings is a policy effect and not a sampling artefact. This also makes the objective a deterministic function of the policy, which is what licenses coordinate ascent.
- **Deviation profiles.** Before any structure in a solved schedule is described in words, each element is reset to a neutral reference and the cost measured in basis points. A solved glide path can look highly structured while most of its structure is worth nothing; the profile separates the two.
- **Grid-edge checks.** An optimum sitting on the boundary of its own search grid is a truncation, not an optimum. Every such case is flagged in the text and, where it mattered, the grid was widened and the analysis rerun.

These are not stylistic preferences. Section 28 of the companion study reports a result that fell by roughly sixty percent when a matched baseline replaced an unmatched one, Section 19 of the companion study reports apparent glide-path structure that the deviation profile dissolved, and Section 27 of the companion study reports a functional-form ranking that reversed its interpretation once the grid-edge check was applied.

5. Baseline Results

Before the pension can be shown to reverse the result, the result has to be reproduced. This section is the replication: the all-equity portfolio against the age-declining glide path, under the earnings-related schedule the original study assumes.

All results in this section use 100,000 simulated lifetimes per strategy, drawn from the same bootstrap sample so that strategies face identical market histories. Certainty equivalents are over the retirement window, for the reason given in Section 4.3.

5.1 The headline comparison

Table 8. Headline lifecycle outcomes by strategy

Strategy	CEC γ=2	CEC γ=5	CEC γ=10	P(ruin)	Median wealth at 63	Median cons.	5th pct cons.	Median bequest
50/50 domestic/international equity	1.406	1.048	0.756	11.7%	21.2	1.483	0.707	44.7
100% domestic equity	1.183	0.882	0.668	25.4%	16.1	1.241	0.579	16.7
100% international equity	1.521	1.108	0.782	9.0%	24.4	1.613	0.755	64.2
60/40 domestic equity/bonds	1.105	0.871	0.671	24.2%	13.3	1.149	0.585	10.2
Target-date fund (glide path)	1.214	0.943	0.704	22.2%	16.4	1.273	0.630	9.4
100% bills (cash)	0.847	0.739	0.618	56.4%	6.4	0.856	0.521	0.0

Notes. Consumption and wealth are expressed as multiples of the investor's real income at age 25. P(ruin) is the probability of exhausting the portfolio with retirement years still to fund. 100,000 paths per strategy on common bootstrap draws.

The ordering is unambiguous at every risk aversion tested. At γ = 5 the 50/50 all-equity portfolio delivers a certainty equivalent of 1.048 against 0.943 for the target-date fund, an advantage of 11.2%. Against the 60/40 portfolio the advantage is 20.3%, and against bills 41.8%.

What makes the result more than a restatement of the equity premium is the tail behaviour. The all-equity portfolio has a *lower* ruin probability (11.7%) than the target-date fund (22.2%), a higher fifth-percentile retirement consumption (0.707 against 0.630), and a lower probability of falling below a seventy-percent replacement target (13.6% against 22.6%). The conservative portfolio is not buying downside protection on this panel. It is paying for the appearance of it.

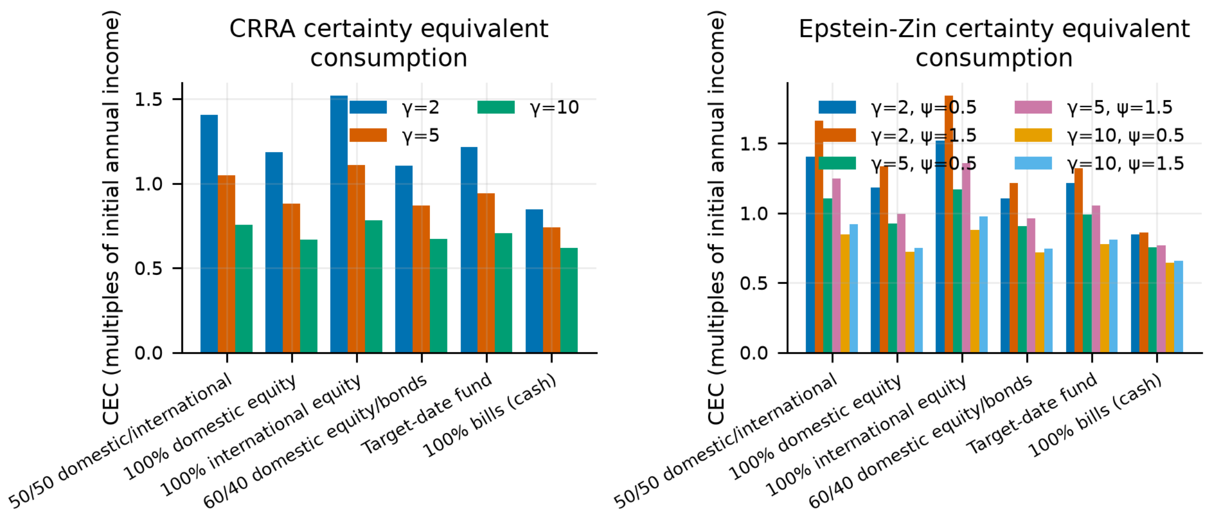

Figure 5.

Certainty-equivalent consumption by strategy and risk aversion. The ranking is stable in γ; the gap narrows as risk aversion rises but does not close within the range tested.

5.2 The mechanism is international, not equity

Domestic equity alone is *not* the winning strategy. Its certainty equivalent at $\gamma = 5$ is 0.882, below the 50/50 portfolio's 1.048, and its ruin probability is 25.4% — nearly double the diversified portfolio's. Pure international equity does better still (1.108), which is a consequence of the leave-one-out construction rather than a recommendation: an investor holding "international" equity in this model holds a 15-country average, and no single real investor has that opportunity set without also holding their own market.

This is the one place the replication does not reproduce ACO's headline. Their recommended portfolio is an even 50/50 split, held for life; on this panel that split is beaten by the international leg alone. We read the difference as a property of a 16-country panel rather than a correction to them. The domestic leg here is a single draw from a set that includes several century-long underperformers, while the international leg averages away exactly that risk; with thirty-nine countries and a tradeable international index the two legs are far closer in character, and the case for holding both is correspondingly stronger. Nothing measurable here contradicts the 50/50 recommendation — the narrower claim this panel supports is that a diversified sleeve dominates a concentrated one.

The practical reading is that the case for equities here is a case for *diversified* equities. A single national market can and did deliver multi-decade real losses; the cross-section of national markets did not do so simultaneously. This is a claim about the covariance structure of the panel, and it is the reason the exercise needs an international sample rather than a longer US one.

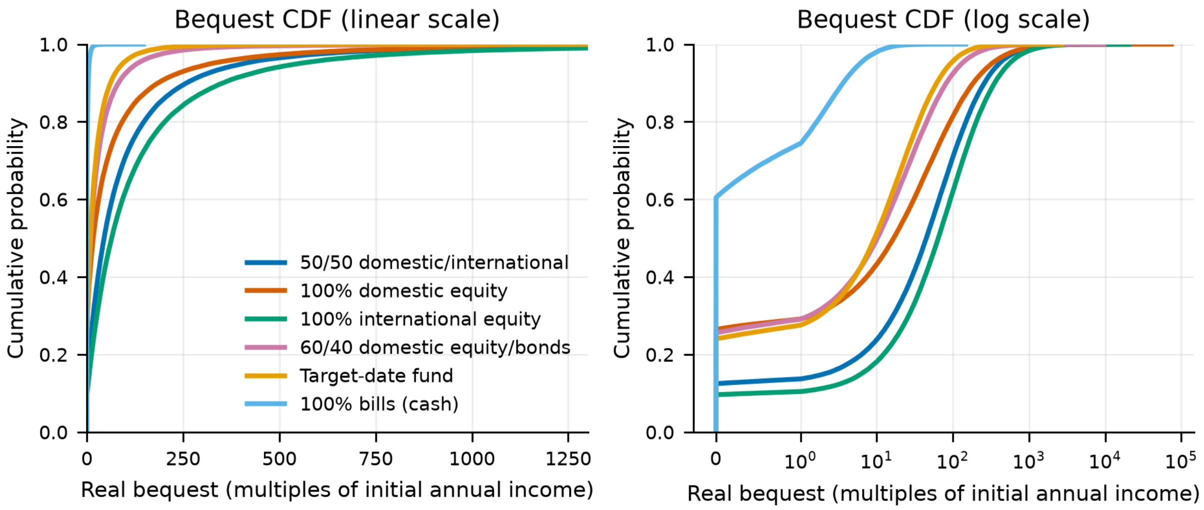

Figure 6.

Cumulative distribution of wealth at retirement. The all-equity distributions stochastically dominate the conservative ones over almost the entire range, including the left tail where the conservative case is usually made.

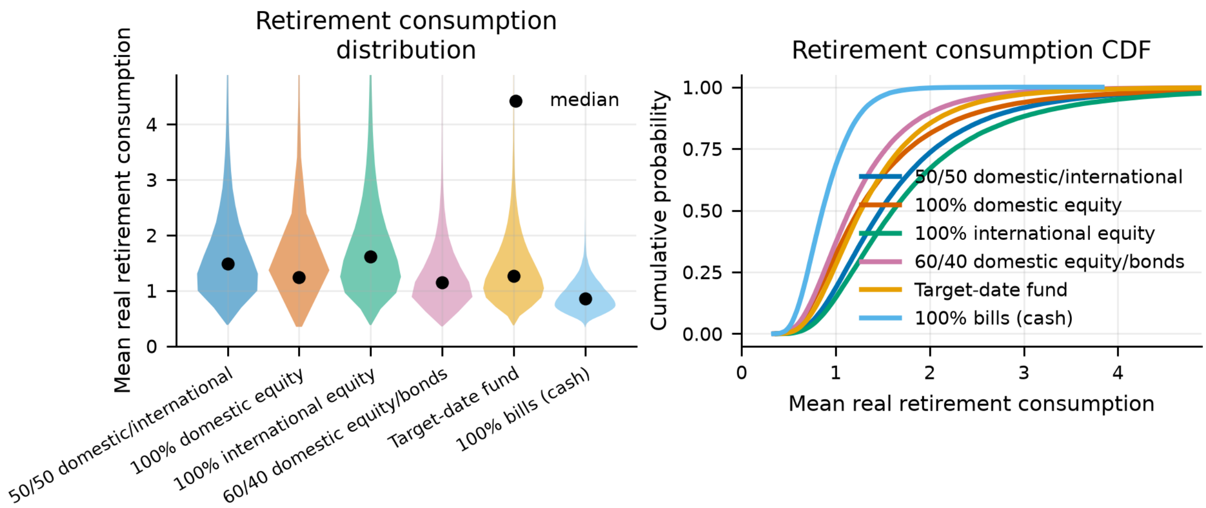

Figure 7.

Distribution of average real retirement consumption by strategy. The vertical line marks the seventy-percent replacement target used in the shortfall statistics.

5.3 Sustainable withdrawal rates

The four-percent rule is the most widely used number in retirement planning. It comes from US historical sequences and it does not survive this panel.

Table 9. Sustainable withdrawal rates on the international panel

Strategy	Sustainable rate at 5% ruin	Ruin probability at 4%
100% international equity	3.48%	9.1%
50/50 domestic/international equity	3.17%	11.7%
Target-date fund (glide path)	2.10%	22.0%
60/40 domestic equity/bonds	1.69%	24.0%
100% domestic equity	1.57%	25.2%
100% bills (cash)	1.07%	56.0%

Notes. The sustainable rate is the highest initial real withdrawal rate, inflation-adjusted thereafter, that leaves at most a five-percent probability of exhausting the portfolio over a 30-year retirement.

At a five-percent ruin tolerance the sustainable rate is 3.17% for the all-equity portfolio and 2.10% for the target-date fund. Withdrawing four percent produces a ruin probability of 11.7% and 22.0% respectively. Two things follow. First, the ordering of strategies is the same here as everywhere else in the paper — the equity portfolio sustains a higher rate, not a lower one. Second, the level is a serious warning: on a sample that includes the twentieth century as most countries lived it, four percent is roughly a one-in-seven proposition even on the best strategy tested.

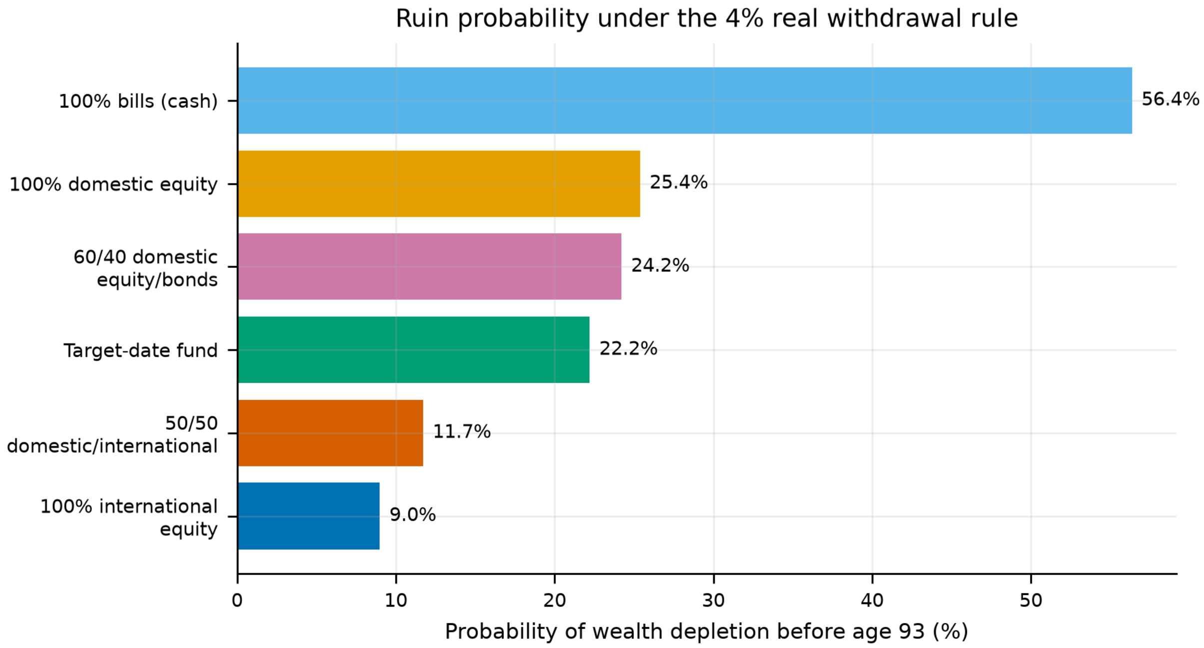

Figure 8.

Probability of portfolio exhaustion as a function of the initial withdrawal rate. The horizontal line is the five-percent tolerance used to define the sustainable rate.

6. What One Country's Pension Does to the Result

This is where the paper starts rather than where it ends, and it is worth saying which of the two before the numbers arrive. Nothing changes but the pension the household retires onto, and the ordering the previous section established reverses. Two things qualify that, both established later and both worth carrying through this section: the reversal appears under one of the eight withdrawal rules this paper runs, and it is the one cell of that table whose sign sixteen countries cannot resolve. Neither is a reason to skip the section -- the reversal is what the literature's standard assumption produces, and understanding why it happens is what the rest of the paper is built on -- but every number below should be read as a point estimate with the interval reported in the ordering section attached.

Every result to this point pays the same public pension in all 16 markets: the US primary-insurance-amount schedule, progressive in career earnings and paid in full whatever else the retiree owns. It is the schedule of exactly one country in the panel, and the panel's whole point is that countries differ.

It is also the shape most flattering to the argument. An earnings-related pension paid regardless of wealth is a **risk-free real annuity**: it arrives whatever the portfolio does, which puts a floor under retirement consumption, crowds fixed income out of the financial portfolio and pushes the optimal equity share up. Nothing in this paper has tested how much of the result that floor is carrying.

Australia is the developed world's counter-example, and it differs in two ways at once. The **pension** is flat rather than earnings-related and is **means-tested**: assessable assets above a free area withdraw it on a taper until it cuts out. And the **contribution** is compulsory: the Superannuation Guarantee pays 12% of ordinary time earnings into a separate fund, on top of whatever the worker saves voluntarily. The two push in opposite directions, so the sweep crosses them rather than running one Australia-versus-America row that would confound them.

6.1 Calibration

Table 10. The Age Pension for a single retiree, March 2026 indexation, with assets-test thresholds from July 2026, against Australian average weekly ordinary time earnings of $2,051.10 (ABS, November 2025).

Quantity	Statutory	In average earnings
Maximum single rate	$1,200.90 a fortnight	0.293
Assets-test free area (homeowner)	$321,500	3.014
Assets-test free area (non-homeowner)	$600,000	5.625
Taper	$3 a fortnight per $1,000	0.078 a year
Cut-out (homeowner)	$722,000	6.767
Superannuation Guarantee	12% of earnings, taxed 15% on entry	0.102 reaching the fund

Notes. Rates are held as multiples of economy-wide average earnings so the schedule travels across the panel's currencies unchanged.

The guarantee is a second contribution stream, not a larger first one. The worker still saves this paper's own 10% out of take-home pay; the employer pays 12% on top, 15% of it is taken as contributions tax, and the remaining 10.2% is invested in the same strategy and assessed by the same means test. Total contributions are 20.2% of income against the American saver's 10%. Because both pots hold the same strategy and face no differential tax on earnings in this model, they are financially one pot and are simulated as one; the guarantee supplies 50% of everything contributed.

Working-life consumption is unchanged by the guarantee here, because its statutory incidence is on the employer. If the true incidence is on workers through lower wages — which is what most of the empirical literature finds — an Australian is paying for it in forgone pay and the comparison below is generous to them by that amount. It does not touch the certainty equivalents, because the utility window in this paper is retirement only. It does mean this is a comparison of *systems as legislated* rather than of two workers with the same lifetime resources, which is why the matched-contribution rows are there.

The taper is the other mechanism, and it is steep. A dollar of assessable assets inside the tapered band costs 7.8% of pension every year it is held. No asset in this panel earns that reliably in real terms, so inside the band the marginal return on wealth is negative — and run backwards, a retiree whose portfolio falls is met by a pension that rises. A means test is a floor and a ceiling at once, which is why it is assessed on assets as they stand, year by year, rather than settled once at retirement.

6.2 Crossing the two features

Table 11. Retirement consumption relative to the US baseline, γ = 5, 50,000 lifetimes per regime.

System	Certainty equivalent (%)	Mean consumption (%)	5th percentile (%)	All-intl over 50/50 (%)	Best strategy
US Social Security (the paper's baseline)	+0.0	+0.0	+0.0	+5.73	100% international equity
US Social Security, saving 20.2%	+23.1	+63.1	+32.8	+6.98	100% international equity
Age Pension rate, paid to everyone (no means test)	-12.9	-12.1	-10.1	+6.33	100% international equity
Australian Age Pension, saving 10%	-49.9	-34.5	-51.6	-8.06	100% bills (cash)
Australia as legislated: 10% voluntary + 12% SG, means-tested pension	-41.0	+26.8	-26.5	-6.42	Target-date fund
Australia as legislated, non-homeowner thresholds	-34.3	+27.7	-20.3	-5.28	Target-date fund

Notes. The first three columns are levels — how much retirement consumption a system delivers. The fifth is the ranking question — which portfolio it leads its investor to hold. They are different questions and here they have different answers.

The Australian system raises the average and lowers the certainty equivalent. As legislated it delivers +26.8% mean retirement consumption against the American baseline and -41.0% certainty-equivalent consumption. Both are right, and the distance between them is the whole result: the fifth percentile is -26.5%. Compulsory saving buys a bigger portfolio and the means test takes away the floor that portfolio would otherwise sit on, so the good outcomes get better and the bad ones get worse. A criterion as averse to the lower tail as a certainty equivalent at γ = 5 prefers the annuity.

The crux is one number. Under the American schedule this investor collects a public pension worth 44% of economy-wide average earnings, every year of retirement, unconditionally and for life. The Australian maximum rate is 29% to begin with — and this investor collects a mean of 2.1%, because the guarantee has made them wealthy enough that the assets test withdraws almost all of it. They have swapped a large risk-free real annuity for a larger portfolio. That is a good trade on the average and a bad one in the tail, which is exactly what the three level columns above report.

The crossing says where each piece of that comes from. Holding the American pension and paying the Australian contribution rate is worth +23.1%: compulsory saving on its own is a large gain. Holding the contribution rate and cutting the pension to the Age Pension's flat rate, still paid to everybody, costs -12.9% and changes nothing about the ordering — it is a smaller annuity, not a different kind of one. Adding the assets test to that is what does the damage. Together they land at -41.0%: the guarantee does not buy back what the means test removes.

6.3 What it does to the ranking

The point estimate reverses the ranking. All-international leads the 50/50 split by 5.73% under the American schedule and by -6.42% under the Australian one, where the best strategy becomes *Target-date fund*. This is the only place in this paper where the headline ordering fails, and it fails for a reason that has nothing to do with the return panel. How precisely sixteen countries can resolve that failure is a separate question, and Section 10 answers it: not at all. The sign reported here is a point estimate, and every statement about it in this section should be read with that section's interval attached.

The mechanism is the floor, not the taper. The taper is certainly steep — 7.8% of assessable assets a year, more than domestic equity earns on average in this panel — and it is tempting to read the reordering off it directly. That reading does not survive checking where this household stands. Section 7.1 measures it: under either reading of the contribution rate the household is past the cut-off before the test is applied, so the taper never operates on them in the central case. It reaches them only in the left tail, once a portfolio has already fallen far enough to qualify.

What the means test removes is the American schedule's unconditional annuity, and with it the floor under bad outcomes. A retiree standing on a floor can afford the equity tail; a retiree standing on their portfolio alone cannot, and a risk-averse objective prices that difference heavily. It shows in the numbers running in opposite directions at once: against the American baseline this household's mean retirement consumption is +26.8% while its fifth percentile is -26.5%. The control is the untested row above — pay the same flat Age Pension to everyone, means-testing nothing, and the ranking is unchanged even though the level falls. Nothing about the returns has changed between any of these rows. What changed is whether anything is guaranteed.

How far it goes depends on the balance, not the country. Run the same means test on a saver with no guarantee behind them — 10% voluntary saving and nothing else — and the lead falls further, to -8.06%, with *100% bills (cash)* taking first place rather than *Target-date fund*. The poorer the saver, the deeper into the taper they sit and the further the distortion goes: at 20.2% of income the answer is a de-risking schedule, at 10% it is cash. The compulsory contribution does not remove the distortion, it softens it.

Two things follow, and they should be kept apart. The first is about this model: its investor contributes 20.2% of a rising income for 38 years and compounds it at historical real equity returns with no tax on fund earnings, which makes them far wealthier than a median Australian and puts them at the top of the taper rather than in the middle of it. The second is about the paper as a whole: every other section models a retiree whose public income is an unconditional annuity, and a retiree whose public income is withdrawn at the margin is solving a different problem. The answer here is not that Australians should hold a glide path. It is that a means test changes what the portfolio is for.

The assets test has a higher free area for retirees who do not own a home, and the model's investor owns nothing outside the portfolio. Running the non-homeowner thresholds — a free area of 5.63 times average earnings rather than 3.01 — moves the certainty equivalent to -34.3%, which is enough to matter and is reported alongside.

Table 12. Where retirees sit on the taper under the system as legislated, holding the 50/50 portfolio.

Age	Full rate	Part rate	No pension	Mean pension (× average earnings)
70	0.8%	4.2%	95.0%	0.008
80	5.2%	5.3%	89.5%	0.023
90	13.2%	3.8%	83.0%	0.045

Notes. A means test that never binds is a flat pension in disguise; one that always binds to zero is no pension at all. This table says which of the two this saver faces.

Table 13. How much of retirement the state is paying for.

System	Mean pension	Share of retirement consumption
Age Pension rate, paid to everyone (no means test)	0.449	29.7%
Australian Age Pension, saving 10%	0.063	5.6%
Australia as legislated: 10% voluntary + 12% SG, means-tested pension	0.033	1.5%
Australia as legislated, non-homeowner thresholds	0.048	2.2%

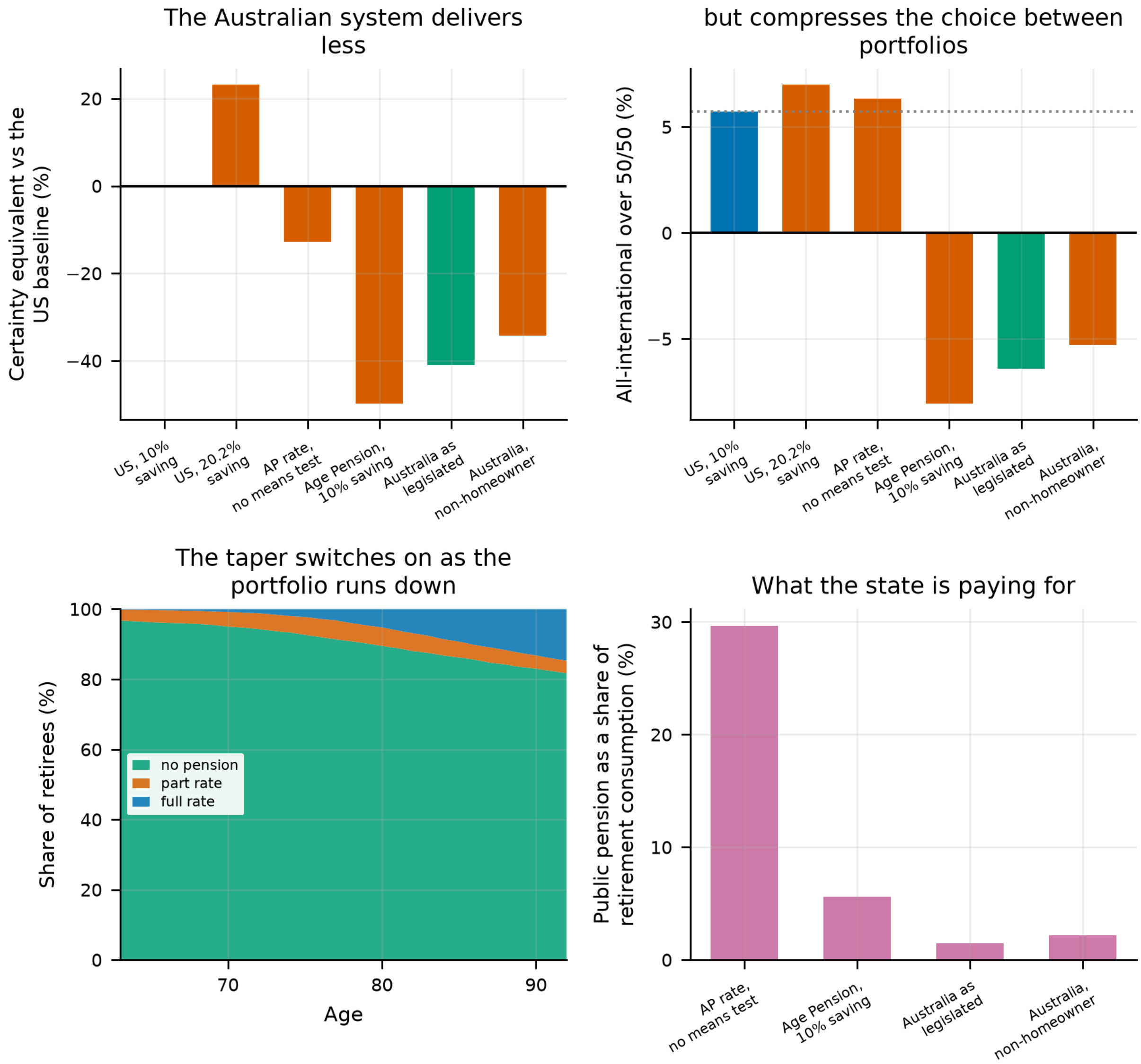

Figure 9.

Top left: retirement consumption under each regime against the American baseline. Top right: what each regime does to the choice between portfolios. Bottom left: where retirees sit on the assets-test taper as the portfolio draws down. Bottom right: how much of retirement consumption the public pension supplies.

6.4 What this changes

- **A pension schedule is not an innocent modelling choice.** Swapping one developed country's for another's moves certainty-equivalent retirement consumption by -41% while the return panel, the preferences and the portfolios stay exactly where they were. Every level quoted elsewhere in this paper is conditional on the American schedule.
- **The average and the certainty equivalent disagree, and both are true.** Compulsory saving raises mean retirement consumption +27% and lowers the fifth percentile -27%. A reader who cares about expected consumption should read the first; one who cares about the bad case should read the second. The paper's own criterion reads the second.
- **The allocation ranking reverses — as a point estimate.** This is the only place in this paper where it does, and Section 10 shows it is also the one cell of that section's table whose sign this cross-section cannot resolve. Under the means test the best strategy becomes *Target-date fund*, and for a saver poorer than this one it goes further still. Every other section models a retiree whose public income arrives regardless of what they own; that is the assumption the ranking rests on.
- **What is not modelled**: the income test, the family home’s exemption from the assets test — the largest single feature of the real system — the tax on fund earnings in accumulation, and every other tax in this paper. Nor is an annuity, which is what the American schedule is and what an Australian retiree would have to buy to match it.

7. Which Feature of the Pension Does the Work

The point estimate has to be attributed to something, and there are two candidates: the means test's taper, and the loss of the unconditional floor beneath it. Australia's pension differs from the American schedule in two ways at once -- it is worked out by a means test rather than from a career of earnings, and it starts on a fixed birthday rather than when work stops -- so this section crosses the two features and reports each alone against both together. Because the second feature is about timing, the retirement date is re-solved in every cell rather than held fixed, which is what the date columns in the table report and why these certainty equivalents are aggregated from the start of working life. They are not comparable with the levels in the previous section; only the differences within this one should be read.

7.1 Which of the two differences does the work

Table 14. Each feature of Australia's pension against the American baseline, γ = 5.

What changes	Best date	Years later	CEC	vs baseline
when the benefit starts	58	-2	0.7754	+0.0086
how the benefit is worked out	67	+7	0.6292	-0.1376
both together	67	+7	0.6292	-0.1376
interaction	—	+2	—	-0.0086

Notes. The baseline is the American schedule: earnings-related, payable from retirement, actuarially reduced for claiming early.

It is how the benefit is worked out, not when it starts. Changing only the start date moves the best date by -2 years. Changing only the formula — a means-tested pension replacing an earnings-related one, still payable the day work stops — moves it +7. The eligibility gate moves the date the *opposite* way to the system it belongs to, and the reason is worth stating: a pension payable at a fixed age is not reduced for stopping early, whereas an actuarially adjusted one is. The gate removes a bridge and a penalty at once, and on this panel the penalty is worth more than the bridge.

The two do not add. The interaction is +2 years against a joint +7, or 29% of it — which is what one feature disarming the other looks like. Once the means test has taken the pension away, the birthday it would have arrived on stops mattering.

And the formula does so much because this household is not poor enough to be paid. The assets test begins withdrawing the pension at 3.0× average earnings and reaches zero at 6.8×. The household in the 2×2 above — which holds contributions fixed, and so saves only what this paper's own saver does — has a median of 13.8×, 2.0 times the cut-off, with 83% of them past it. Against a full rate worth 29% of career-average income, what is actually paid averages 3.9%; the American schedule, means-tested against nothing, pays 43%. So this is only partly a comparison of pension *designs*. It is substantially a comparison between a household that receives a pension and the same household receiving almost none.

That household is not, however, an Australian one. Holding contributions fixed is what separates the pension's timing from its formula, and it is the wrong household for any sentence about Australia, which makes its contributions compulsory. Carrying the Superannuation Guarantee moves the median from 2.0 to 4.7 times the cut-off, the share past it from 83% to 98%, and what the pension actually replaces from 3.9% down to 1.9% of career income. The guarantee does not carry this household *through* the means test; it carries them 2.3 times further past it.

This settles what the taper can and cannot be blamed for. It is steep enough to act as a wealth tax — every dollar inside the band costs more pension a year than domestic equity earns in this panel — but a rate that is never reached cannot be the mechanism behind anything. Under either reading the household is past the cut-off before the test is applied, and the taper touches them only in the left tail, once a portfolio has already fallen far enough to qualify. What separates the two systems here is therefore not the withdrawal rate on the pension. It is that one system pays this household an unconditional annuity and the other pays them almost nothing, and the annuity is worth most exactly where the portfolio is worth least.

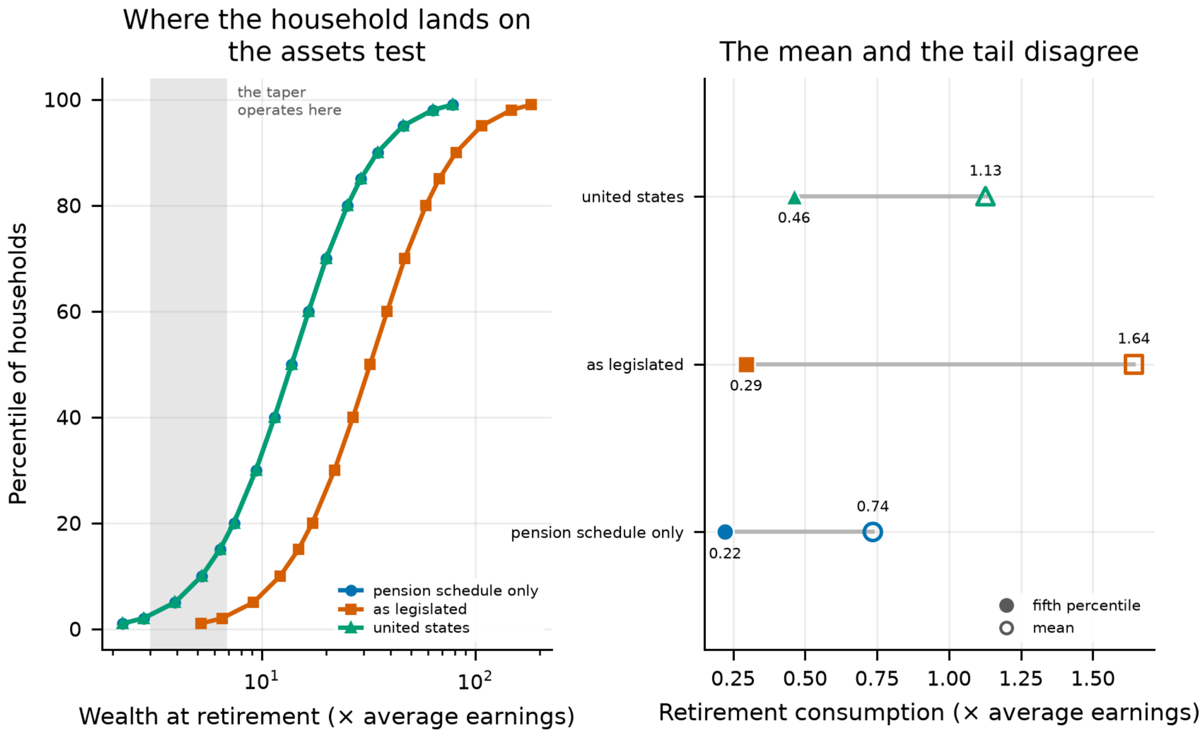

Figure 10.

What the assets test does, and does not, explain. Left, the distribution of wealth at retirement for each household against the band the taper operates in — on a log scale, because the distance involved is multiplicative. Both Australian households sit almost entirely to the right of the band, so the taper is answered before it is applied. Right, the mean and the fifth percentile of retirement consumption for each. The two run in opposite directions: the household with compulsory saving has the highest mean of the three and nearly the lowest tail, which is the gap the certainty equivalent is pricing.

7.2 The withdrawal rule is half the comparison

Everything above spends by this project's own rule: a fixed real amount set as a share of the portfolio at retirement. That rule cannot answer the question a pension is *for*. It sets the standard of living from the portfolio, so a larger balance spends more rather than lasting longer, and the pension is spent on top of the withdrawal rather than funding part of it. The alternative fixes the target instead — a set share of pre-retirement income for life, the pension netted off, the portfolio paying the rest.

Table 15. Each pension system under each withdrawal rule, γ = 5, retiring at 63.

Rule	Pension system	CEC	Ruin (%)	Mean c	5th pct c
fixed real	US social security	1.0494	11.7	1.729	0.708
replace 50	US social security	0.6057	2.5	0.852	0.438
replace 75	US social security	0.6759	11.7	1.136	0.454
replace 100	US social security	0.7578	24.8	1.373	0.500
fixed real	Age Pension only	0.4226	11.7	1.129	0.341
replace 50	Age Pension only	0.3417	10.3	0.755	0.255
replace 75	Age Pension only	0.4917	26.1	1.014	0.379
replace 100	Age Pension only	0.5782	41.0	1.190	0.449
fixed real	Age Pension + super guarantee	0.7242	11.7	2.520	0.452
replace 50	Age Pension + super guarantee	0.3354	1.9	0.797	0.250
replace 75	Age Pension + super guarantee	0.4991	6.8	1.162	0.374
replace 100	Age Pension + super guarantee	0.6403	13.4	1.485	0.449

Notes. The replacement rules target a share of average income over the final working years, held in real terms.

Under the portfolio rule all three systems ruin at the same rate, to the last decimal. That is an artefact, not a finding, and it is worth naming because this project has leaned on that rule throughout: withdrawals set as a share of wealth at retirement scale with the portfolio, so doubling the balance doubles the spending and exhausts it in the same year. Ruin under that rule is a property of the return sequence alone, and says nothing about how much was saved.

Hold the target still and the Superannuation Guarantee finally shows up where it should. At 100% replacement the Australian household runs out 11.4 percentage points less often than the American one. Compulsory saving buys safety — but only under a rule that permits safety to be bought.

And the certainty equivalent still favours the American schedule at every target. Both facts hold at once, and the combination is the result: the Australian household runs out *less often* and is *poorer when it does*. What remains once the portfolio is gone is a flat means-tested pension rather than an earnings-related one, and a CRRA aggregator at this risk aversion prices the depth of the floor above the frequency of reaching it. A reader who cares about the probability of running out and a reader who cares about the worst case will rank these two systems differently, and neither is misreading the table.

8. Who Pays, and Whom the Test Binds

Two objections stand against everything above, and they look like one. The first is that the Superannuation Guarantee has been free: its statutory incidence is on the employer, so the Australian arm has been handed a tenth of income the American arm never got, and most of the empirical literature puts the economic incidence on wages instead. The second is that the household simulated here retires far past the assets test -- on 36.4 times average earnings at the all-equity portfolio this sweep holds, against a cut-off of 6.8 -- so the means test at the centre of the paper never binds it. (Balances differ a little between sections because the portfolio differs; every one of them is several times the cut-off, which is the only part that matters for the objection.)

The tempting move is to answer both at once: charge the guarantee, watch the balance fall, and let the household walk down onto the test. It does not work, and the reason is the cleanest thing in this section. Under full economic incidence the worker funds the contribution out of wages, so take-home pay falls; but the contribution is still made, so the fund receives the same money and compounds it identically. **Incidence changes what the household gave up, not what it arrives with.** The balance at the pension age is the same to the last cent at either end of the dial, and the two objections need two answers.

8.1 Charging the guarantee

The first dial charges a share of the employer contribution against wages, from nothing to all of it, crossed with every allocation: 105 combinations on common random numbers. Charging it in full costs 13.4% of lifetime certainty-equivalent consumption and lowers mean consumption during the working years by 14.8%, which is the transfer itself. That is the size of the free lunch, and the honest reading of every cross-system comparison above is that it is bracketed by the two ends of this dial rather than pinned at either.

Measured the way the rest of this paper measures — a certainty equivalent over the retirement window — the answer is exactly zero, 0.7701 at both ends to every digit. The retiree's problem is untouched by who paid for the balance: same wealth, same pension, same returns. And the allocation does not move either: 100% equity is optimal whether the guarantee is free or funded out of forty years of wages. What the free guarantee inflated was the level of Australian consumption, not the portfolio chosen to deliver it.

8.2 A household the test does bind

The second dial moves the balance directly. It scales what reaches the pension age, applied at the retirement boundary and nowhere else, so the career is identical across the grid and a change in the retiree's allocation is a response to the balance rather than to what was given up for it. It is a counterfactual and worth being blunt about: not a claim that Australians save less than this household, but the observation that a household near the assets test holds a fraction of what this one holds, and that the retiree's problem at that balance is the one the model in Section 2 speaks to. Nothing near the test is reachable otherwise.

Table 16. The equity share a retiree wants, by where the balance leaves them against the assets test, under two withdrawal rules. Balances are multiples of economy-wide average earnings; the free area is 3.01 and the cut-off 6.77. The pension begins the day work stops in both columns, so every retirement year is under the test.

Median balance at retirement	Position	Wanted equity, fixed real rule	Wanted equity, share of balance
0.72×	below the free area	0%	100%
1.45×	below the free area	0%	100%
2.42×	below the free area	0%	100%
3.38×	inside the taper band	0%	100%
4.35×	inside the taper band	0%	100%
5.31×	inside the taper band	0%	100%

Median balance at retirement	Position	Wanted equity, fixed real rule	Wanted equity, share of balance
6.28×	inside the taper band	0%	100%
7.25×	above the cut-off	0%	100%
8.21×	above the cut-off	0%	100%
9.66×	above the cut-off	0%	100%
12.08×	above the cut-off	0%	100%
16.91×	above the cut-off	0%	100%
28.98×	above the cut-off	10%	100%
48.30×	above the cut-off	55%	100%

Notes. Each row is the best of twenty-one equity shares at that balance, scored on certainty-equivalent retirement consumption over the same simulated lifetimes.

Under a fixed real withdrawal the prediction fails, and it fails in the direction the wealth-tax reading would have picked. Wanted equity is 0% below the free area, 0% inside the band and 55% above the cut-off — monotone in wealth, with the minimum where the model said the maximum should be. A household the means test actually binds wants no equity at all.

The reason is the withdrawal rule, and it is worth stating precisely because it is the most useful thing in this section. A fixed real rule spends a set amount and never spends what the portfolio earns. For a household near the assets test that is close to the worst available arrangement: a good equity outcome converts into assessable assets, the pension is withdrawn against them at 7.8% a year, and consumption does not rise by a cent. The upside is confiscated and the downside is not. Equity is then dominated, and no amount of insurance in the shape of the budget line can rescue it.

Change the rule and the answer changes completely. Spending a share of the *current* balance instead — so the gain is consumed as it arrives rather than assessed — the same household wants 100% equity at every position, below the free area, inside the band and past the cut-off alike. The swing between the two rules averages 95 percentage points, which is the whole allocation. So the portfolio a means-tested retiree should hold is not a fact about the means test. It is a fact about the means test and the drawdown rule together — and the rule is the half the retiree can choose.

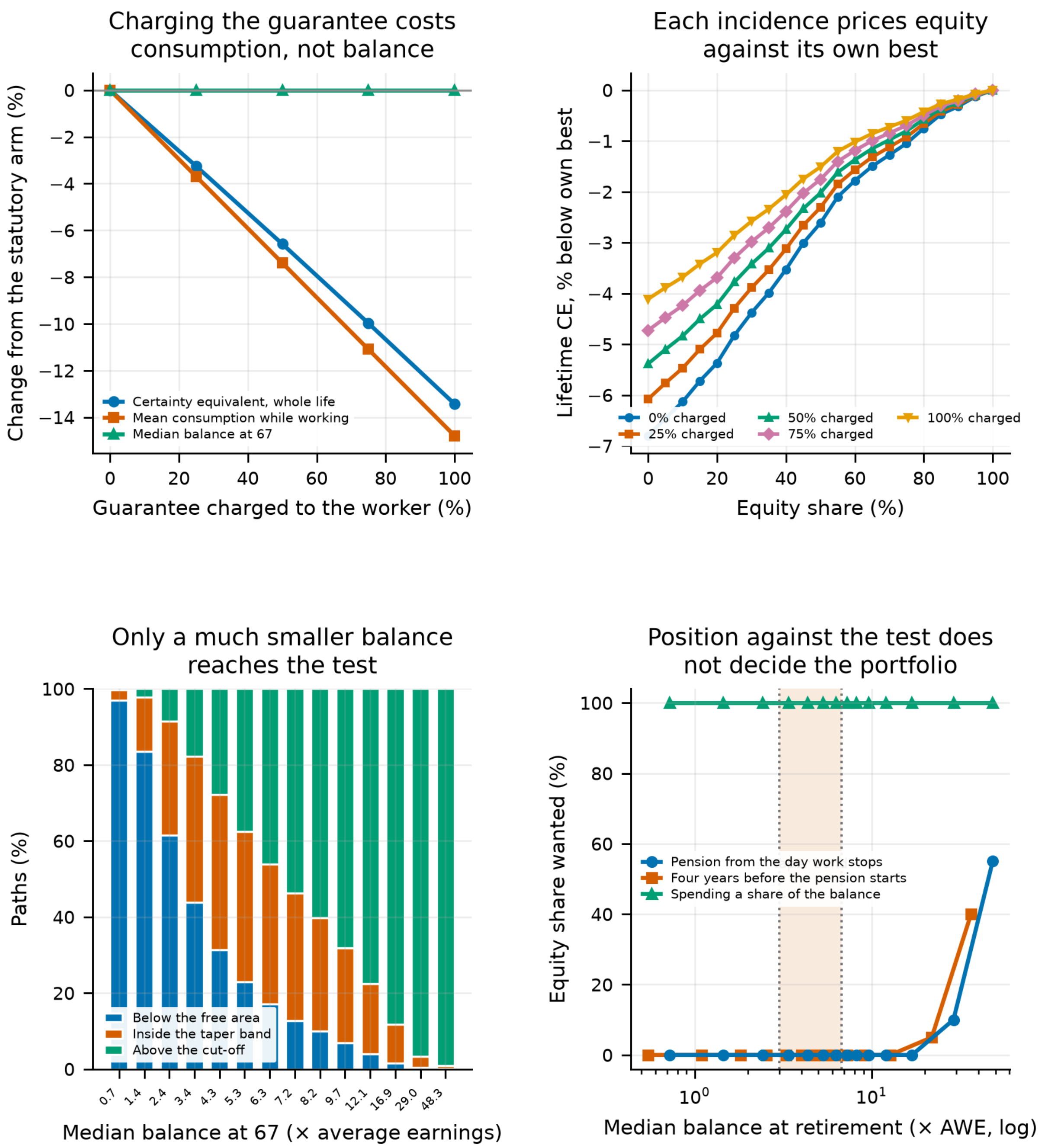

Figure 11.

Two dials on the Australian arm. Charging the Superannuation Guarantee to wages (top row) moves lifetime consumption and leaves the balance untouched; moving the balance (bottom row) is the only way to put a household on the assets test. The lower right panel is the model's prediction, tested.

8.3 What the panel costs this corner

The two results above are checked against the preference specification and not against the panel, which is the objection Section 10 answers for its own numbers. So the balance sweep is recomputed 16 times, once with each country's history removed, and the equity share the retiree wants is read off each one.

The answer is 0% equity in every one of the 16 deletions. Not close to 0% — identical, to every digit the sweep resolves. A jackknife standard error would be exactly zero and the interval a point, which is why we report the deletions rather than an interval computed from them. That is what a corner solution looks like when it is checked: the objective is not nearly best at the boundary, it is pinned there, and no fifteen-country subsample of this panel moves it. It is also the reason to state the finding as a direction rather than a number. The panel resolves *which corner* with no uncertainty at all and says nothing about how far inside the boundary a real retiree, facing a real assets test rather than this stylisation, would land.

This is the same lesson the withdrawal-rule section reaches from the other side, and the two should be read together. There it was a rule that cannot deplete restoring the floor an asset test removed; here it is a rule that cannot spend destroying the case for equity that a floor would otherwise support. Both say the drawdown default and the portfolio default are one decision, and that a plan sponsor choosing either without the other is choosing in the dark.

9. The Withdrawal Rule a Retiree Should Actually Use

If the mechanism is a missing floor rather than a tax on wealth, a household should be able to build its own floor out of the withdrawal rule. Before that can be tested, the rules have to be compared on an honest footing: every comparison so far scores a retirement that ends on a known date, which hands the rules that amortise to one the answer. This section re-solves the rule, the rate and the allocation together against a real survival curve, and finds which rule a retiree should actually use. What that rule then does to the pension comparison is the next section.

Every comparison of withdrawal rules so far — Section 30 of the companion study's sweep, Section 31 of the companion study's joint optimisation — scores a retirement that ends at ninety-three with certainty. That is not neutral between rules. A rule that amortises to a fixed date is handed the answer, because dividing by the years remaining is very nearly optimal when the years remaining are a known constant. A rule that hedges longevity pays a premium against a risk the model has switched off. And ruin counts as failure a portfolio exhausted at ninety-one for an investor who most likely died some years before.

Section 17 of the companion study re-weights the aggregation by a Gompertz survival curve and finds it does not change which *allocation* wins. It also says, in terms, that the treatment is only approximate for the horizon-based rules, “which would themselves change if they knew the mortality table”. This section is that unfinished sentence: the rule and the rate are re-solved alongside the allocation, on one objective.

The grid is 2,280 combinations of allocation, rule and rate, each scored twice on the *same* simulated lifetimes — once over the fixed horizon, once survival-weighted. No path is re-drawn between the two, so a difference between the scores is the aggregation and nothing else. On this calibration the expected age at death is 81.9, so a retiree stopping at 63 should plan for 18.9 years against the 30 assumed throughout. The model was not merely uncertain about the horizon; it was assuming one substantially longer than the median retiree gets.

Table 17. The best combination under each aggregation, γ = 5.

Scored over	Rule	Equity	Domestic	CEC
fixed horizon	gompertz	100%	10%	1.3844
survival-weighted	amortisation (6% assumed return)	100%	10%	1.4232

Notes. Both winners derive their level from a planning horizon and so set no rate of their own.

The horizon changes the rule and nothing else. Freeing the allocation to be re-chosen for a real lifespan is worth +0.00% and freeing the rate +0.00%; freeing the rule is worth +2.81%, which is the whole of the +2.81% available from re-choosing all three, with an interaction of +0.00%. The three decisions do not interact here, and that is worth stating as plainly as an interaction would have been.

The rate optimum is interior: the best rate-setting rule, percentage of balance, wants 8.0% inside a grid running 3.0% to 12.0%, so the curve turns over rather than stopping and the number is a peak rather than the end of the grid.

The assumed return has an interior optimum too. amortisation (6% assumed return) wants 6% inside a grid running 0% to 12%, so over-assuming does begin to cost and the sweep can see where it starts. This is a dial the rule comparison in Section 30 of the companion study never swept: the amortisation rule enters that menu at assumed returns chosen by hand, and the best of those is not the same thing as an optimum.

The rate a rule wants spans 3.5 percentage points. constant percent wants 8.0% and constant real wants 4.5%, across 9 rules that set a rate at all. The tempting explanation is that the rules which cannot run out want the high rates, and it is nearly right but not right: endowment (light smoothing) wants 8.0% and *can* deplete, tying the top of the list. What sets the rate is how far spending scales with the portfolio rather than the ability to run out as such — a lightly smoothed endowment rule is most of a percentage of balance, and is priced like one.

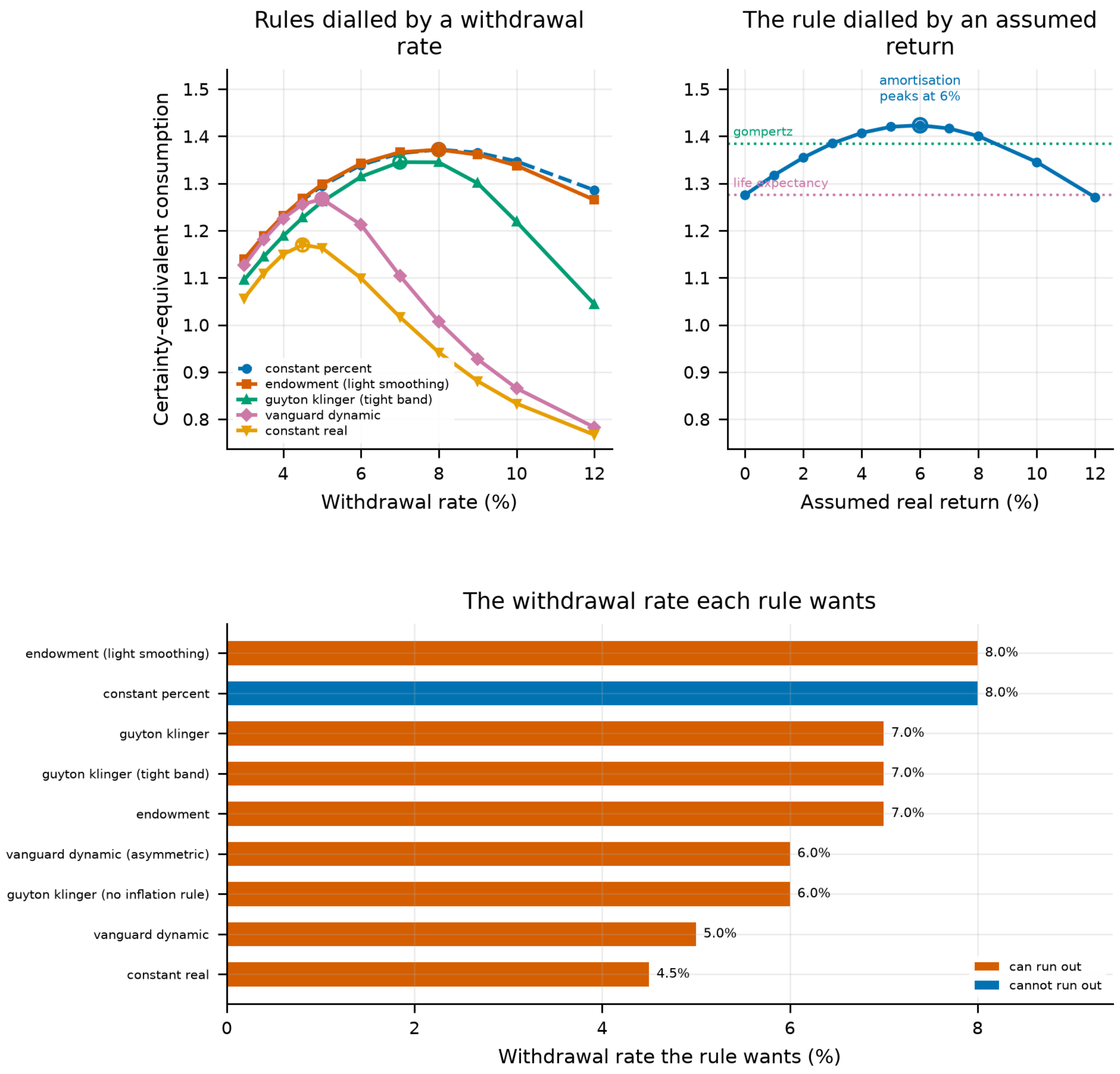

Figure 12.

Where each dial that sets a spending level actually peaks. Top left, the certainty equivalent against the withdrawal rate for the five rules that take one; rings mark where each curve turns over, and solid lines can run out where dashed cannot. Top right, the same against the assumed real return for the rule dialled by one instead, with the two rules that set no dial at all drawn as the constant levels they are. A withdrawal rate and an assumed return are different measures, so they get an x-axis each; the vertical scale is shared between them, which is what lets the two peaks be compared by eye rather than by reading numbers off two ranges. Below, the withdrawal rate each rule wants.

9.1 Which rules a real lifespan promotes

Table 18. Every rule at its own best rate and allocation under the objective being ranked.

Rule	Fixed horizon	Real lifespan	Places gained
amortisation (6% assumed return)	5	1	+4
amortisation (5% assumed return)	3	2	+1
amortisation (7% assumed return)	7	3	+4
amortisation (4% assumed return)	2	4	-2
amortisation (8% assumed return)	11	5	+6
amortisation (3% assumed return)	4	6	-2

Rule	Fixed horizon	Real lifespan	Places gained
gompertz	1	7	-6
constant percent	8	8	+0
endowment (light smoothing)	9	9	+0
endowment	13	10	+3
amortisation (2% assumed return)	6	11	-5
amortisation (10% assumed return)	19	12	+7
guyton klinger (tight band)	14	13	+1
guyton klinger	15	14	+1
guyton klinger (no inflation rule)	18	15	+3
amortisation (1% assumed return)	10	16	-6
gompertz (+5y buffer)	12	17	-5
vanguard dynamic (asymmetric)	22	18	+4
amortisation (0% assumed return)	16	19	-3
life expectancy	16	19	-3
amortisation (12% assumed return)	23	21	+2
vanguard dynamic	21	22	-1
life expectancy (+5y buffer)	20	23	-3
constant real	24	24	+0

Notes. Scoring each rule at a setting chosen to suit a different horizon would confound the rule with the rate.

Part of the reason is the spending level rather than the mortality table. Rank each rule by the share of the balance it draws in the first retirement year — a number every rule has, however it arrives at one — and that order matches the survival-weighted ranking with a rank correlation of +0.48, against +0.06 under the fixed horizon. The contrast is the finding rather than either level, and at +0.48 it is a contributing cause and not the whole of one. The winning rule has never heard of a survival curve: the best rule that reads one ranks 7. So the answer to “should the withdrawal rule know how long you are likely to live?” is that it does not need to — it needs to spend as though the answer were shorter than thirty years, and there is more than one way to arrange that.

9.2 The ruin number everyone quotes

Ruin roughly halves. On the best rule that can actually run out — endowment — the fixed horizon reports 0.9% against the real lifespan's 0.5%, and across all 1,449 combinations that can deplete the median ratio is 2.0. Nothing about any portfolio changed. What changed is that the earlier number counts as a failure a portfolio exhausted at ninety-one for somebody who, on this survival curve, most likely died before reaching it. Every ruin probability elsewhere in this paper carries the same overstatement.

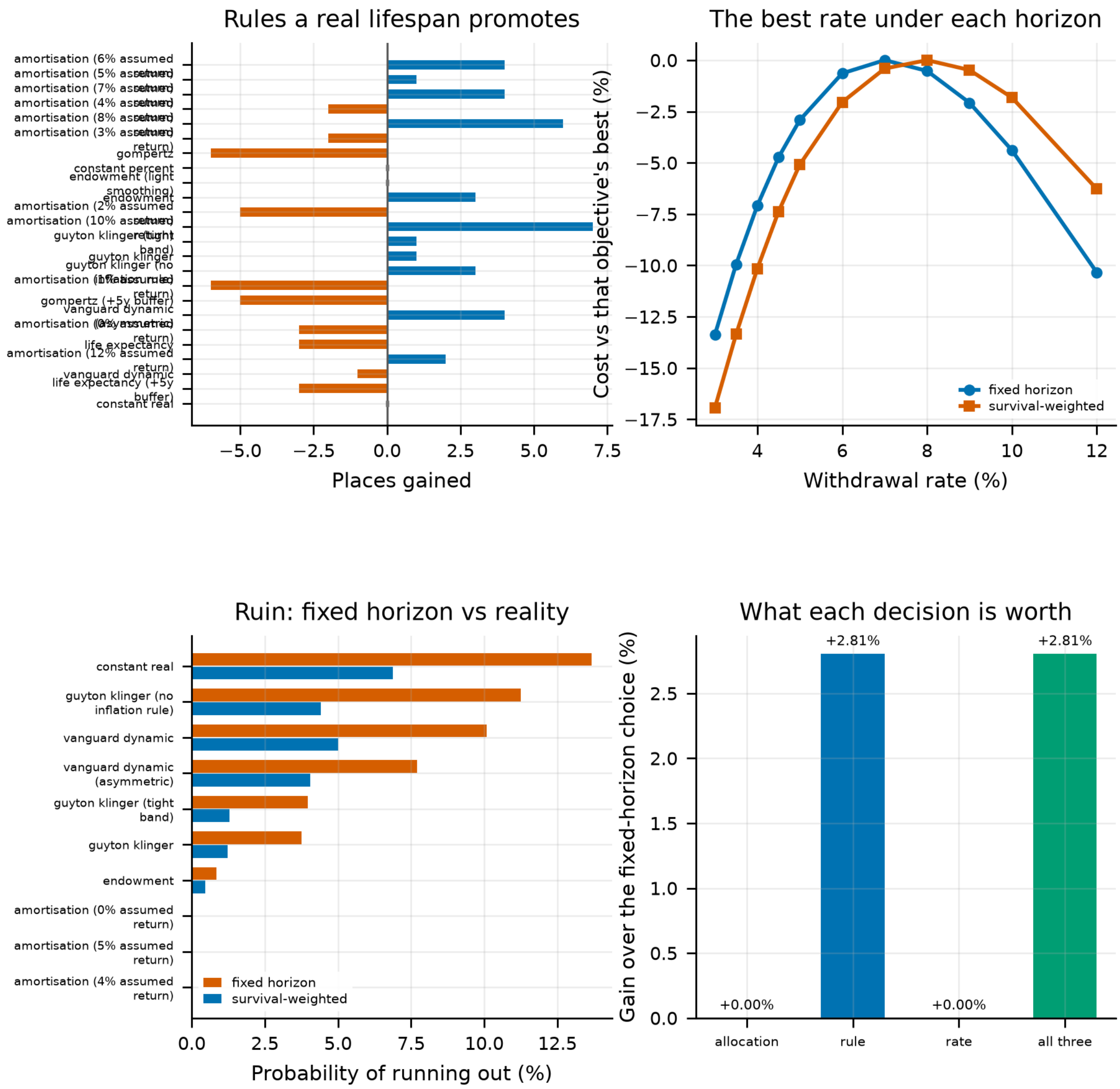

Figure 13.

What an uncertain lifespan does to the rule, the rate and the risk. Top left, how far each rule moves in the ranking; top right, where the best rate sits under each objective, each scaled to its own best so only the shape is compared; bottom left, ruin under both aggregations; bottom right, what re-choosing each decision is worth.

What is not modelled: annuities, which are the direct hedge for the risk this section prices and would dominate part of the grid if they were in it; a rule that conditions on realised health rather than on the table it started with; and couples, where the relevant horizon is the second death and the distribution is a different shape. The Gompertz curve is a two-parameter model rather than a life table, and Section 17 of the companion study sweeps those parameters; the grid here is held at that section's central calibration.

10. Which Portfolio Wins, and Under Which Rule

The previous section found the withdrawal rule a retiree should use. This one asks what that rule does to the question the paper opened with, and it is worth being precise about why the question needs asking separately.

Two orderings have been in play. The first is between *portfolios*: all-equity against the target-date fund, which is what Sections 5 and 6 report and what the target-date literature is about. The second is between *countries*: whether an Australian household consumes more than an American one, which is what a comparison of certainty equivalents across pension systems reports. Section 7 establishes the second under an amortisation rule, and it is tempting to read that as the first reversing. It is not the same quantity, and the two can move in opposite directions. This section reports the portfolio ordering directly.

Table 19. The all-equity portfolio's lead over the target-date fund, in certainty-equivalent retirement consumption, for every pension system and every withdrawal rule. Positive means the all-equity portfolio wins.

Withdrawal rule	United States	Age Pension, no means test	Australia as legislated
fixed real	+11.09%	+12.78%	-2.08%
percentage of balance at 4%	+17.92%	+20.44%	+18.43%
amortisation at 0%	+16.39%	+18.74%	+14.41%
amortisation at 2%	+18.21%	+21.09%	+17.56%
amortisation at 4%	+19.60%	+22.88%	+21.43%
amortisation at 6%	+20.24%	+23.69%	+24.87%
amortisation at 8%	+19.97%	+23.35%	+26.63%
amortisation at 10%	+18.83%	+21.96%	+26.37%

Notes. Every cell is scored on the same simulated lifetimes, so a difference between two cells is the system, the rule or the portfolio and never a different draw of returns. The estate is included, because an amortisation rule spends the portfolio to zero by construction and a fixed real rule does not.

The first finding survives being re-derived on this grid. Under a fixed real withdrawal the all-equity portfolio leads by +11.09% in the American system and -2.08% in the Australian one, on the same regimes and the same path count as Section 6.

These are not the percentages in that section's own table, and the difference is a labelling one worth stating rather than smoothing over. The column headed ‘All-intl over 50/50’ there compares two *all-equity* portfolios with each other — a hundred per cent international against the fifty-fifty domestic-international split — which is a question about the international sleeve, not about equity versus a glide path. Its ranking column carries the portfolio comparison, and agrees with the table above: the target-date fund is first in Australia and is not in the United States. The table above puts a number on it.

And the second finding holds as a statement about portfolios, not only about countries. Under amortisation at 8% the all-equity portfolio leads by +26.63% in the Australian system, against -2.08% under the fixed real rule. 7 of 8 rules in the menu return the lead. A rule that supplies its own floor restores the all-equity prescription, and that is a claim about portfolios rather than an inference from one about countries.

Either way the rule belongs in the statement of the result. Within the Australian system alone, changing nothing but the withdrawal rule moves the all-equity lead by 28.7 percentage points. A plan sponsor choosing a portfolio default without also choosing a drawdown default is choosing on an axis that explains less than the one they left open.

10.1 The reversal lives in the worst cell of the table

One column of the table deserves to be read against the rest, because it carries the paper's headline and it is the only negative entry in 24 cells. The reversal appears under 1 of the 8 rules, and the rule is the fixed real withdrawal — the 4% rule, and the one the lifecycle literature and this paper's earlier sections all spend by. Section 9 has already found that rule is not the one a retiree should use. This table says what it costs the household that does.

Table 20. The all-equity portfolio under each withdrawal rule, in the Australian system. The first row is the rule under which the pension reverses the portfolio ordering.

Withdrawal rule	CEC	Probability of ruin
fixed real	0.6411	11.7%
percentage of balance at 4%	0.8615	0.0%
amortisation at 0%	0.9450	0.0%
amortisation at 2%	1.1329	0.0%
amortisation at 4%	1.2721	0.0%
amortisation at 6%	1.3362	0.0%
amortisation at 8%	1.3217	0.0%
amortisation at 10%	1.2493	0.0%

Notes. Same simulated lifetimes as the table above. Ruin is the share of paths whose portfolio is exhausted before the terminal age.

The fixed real rule leaves this household on 0.6411 against 1.3362 under amortisation at 6% — a reduction of 52% in certainty-equivalent consumption — and exhausts the portfolio on 11.7% of paths against 0.0%. It is dominated on both margins by every other rule in the menu.

That is not a reason to discard the reversal, and we are not discarding it. A fixed real withdrawal is what the withdrawal-rate literature is written about, what most retirement calculators implement, and what the study this paper replicates assumes throughout; a result about the households actually following it is a result about a large number of real people. It is a reason to state the finding with its condition attached. The all-equity prescription survives an asset-tested pension for a retiree who spends a share of the balance, and does not survive it for one who spends a fixed real amount. Which of those a household is doing is a choice, and on this evidence it matters more than the portfolio choice it is usually treated as subordinate to.

10.2 How precisely the panel resolves each sign

Every claim in this paper is a claim about a sign, and the panel is sixteen developed markets whose twentieth centuries were not independent of one another. So each cell above is recomputed sixteen times, once with each country's history removed, and the delete-one jackknife gives the sampling error the panel carries. That is the error to weigh a sign against. Monte Carlo error is not: a hundred thousand paths drive it close to zero without adding a single country of evidence.

Table 21. Delete-one-country intervals on the all-equity lead, for the two systems the headline compares.

Pension system	Withdrawal rule	Lead (%)	Jackknife s.e.	95% interval	Sign holds in all 16
Australia as legislated	fixed real	-2.08	10.35	[-22.35, +18.20]	no
Australia as legislated	percentage of balance at 4%	+18.43	4.76	[+9.10, +27.76]	yes
Australia as legislated	amortisation at 0%	+14.41	3.10	[+8.35, +20.48]	yes
Australia as legislated	amortisation at 2%	+17.56	3.08	[+11.51, +23.60]	yes
Australia as legislated	amortisation at 4%	+21.43	3.27	[+15.01, +27.85]	yes
Australia as legislated	amortisation at 6%	+24.87	3.72	[+17.58, +32.16]	yes
Australia as legislated	amortisation at 8%	+26.63	4.26	[+18.28, +34.98]	yes
Australia as legislated	amortisation at 10%	+26.37	4.66	[+17.24, +35.50]	yes

Pension system	Withdrawal rule	Lead (%)	Jackknife s.e.	95% interval	Sign holds in all 16
United States	fixed real	+11.09	2.35	[+6.49, +15.68]	yes
United States	percentage of balance at 4%	+17.92	3.50	[+11.06, +24.78]	yes
United States	amortisation at 0%	+16.39	2.52	[+11.45, +21.33]	yes
United States	amortisation at 2%	+18.21	2.92	[+12.49, +23.94]	yes
United States	amortisation at 4%	+19.60	3.36	[+13.02, +26.19]	yes
United States	amortisation at 6%	+20.24	3.75	[+12.89, +27.60]	yes
United States	amortisation at 8%	+19.97	3.98	[+12.17, +27.77]	yes
United States	amortisation at 10%	+18.83	3.95	[+11.08, +26.57]	yes

Notes. The jackknife standard error over sixteen sub-panels, each holding fifteen markets. An interval containing zero is a cell in which this panel cannot say which portfolio wins.

The reversal is not a sign this panel can resolve. The contested cell is -2.08% with a delete-one standard error of 10.35, so the interval is [-22.35, +18.20], which contains zero. Across the sixteen sub-panels the cell runs [-3.54, +6.22], and the sign does not hold in all of them.

We therefore state the reversal as what it is: a point estimate this cross-section is too small to separate from zero.

10.3 The interval on the difference, which is the claim

The claim the evidence does support is a difference rather than a level — that changing the withdrawal rule moves the lead by tens of points — and an interval on two levels is not an interval on their difference. The sixteen sub-panels are shared between cells, so the differences are paired and get a jackknife of their own rather than one assembled out of two marginal standard errors.

Table 22. Delete-one-country intervals on the difference between each withdrawal rule and the fixed real rule, in the Australian system.

Rule, against the fixed real rule	Difference (pp)	Jackknife s.e.	95% interval	Cell correlation
percentage of balance at 4%	+20.5	10.1	[+0.8, +40.2]	0.29
amortisation at 0%	+16.5	9.9	[-2.8, +35.8]	0.30
amortisation at 2%	+19.6	9.9	[+0.2, +39.1]	0.29
amortisation at 4%	+23.5	10.0	[+3.9, +43.1]	0.26
amortisation at 6%	+26.9	10.1	[+7.2, +46.7]	0.25
amortisation at 8%	+28.7	10.1	[+8.8, +48.6]	0.25
amortisation at 10%	+28.4	10.2	[+8.6, +48.3]	0.27

Notes. Paired across the sixteen sub-panels. The correlation column says how much the pairing helps: near zero the difference is no better resolved than the levels it is built from.

The widest rule effect is amortisation at 8% at +28.7 percentage points, with a standard error of 10.1 and an interval of [+8.8, +48.6]. 6 of 7 differences exclude zero. The pairing helps less than one might hope: the cells correlate 0.27 across deletions at the median, so these intervals are not much tighter than the levels they come from. We report that too, because the reason to compute a difference interval at all is that it *can* be far tighter, and here it is not.

So the rule effect is resolved and the reversal is not, but by a narrower margin than the point estimates suggest. The honest summary is that sixteen countries can separate a 29-point effect from zero and cannot separate a two-point one — which is what one would expect, and is worth having established rather than assumed.

10.4 And the same grid at other risk aversions

The interval above is what the panel costs in precision. It says nothing about the other objection, which is that a certainty equivalent is a statement about a felicity function as much as about a portfolio. Scoring an outcome again costs nothing next to producing it, so the whole grid is re-scored at risk aversions of 2, 5 and 10. This is the check Section 8 runs against preferences and this section had been running only against the panel; each result is now checked against both.

Table 23. The all-equity portfolio's lead over the target-date fund in the Australian system, at three risk aversions.

Withdrawal rule	γ = 2	γ = 5	γ = 10
fixed real	+21.91	-2.08	-36.40
percentage of balance at 4%	+45.48	+18.43	+7.91
amortisation at 0%	+39.38	+14.41	+5.25
amortisation at 2%	+40.97	+17.56	+8.53
amortisation at 4%	+41.98	+21.43	+16.50
amortisation at 6%	+42.22	+24.87	+21.42
amortisation at 8%	+41.53	+26.63	+20.96
amortisation at 10%	+39.92	+26.37	+18.68

Notes. The same simulated lifetimes throughout: only the objective they are scored under changes, so the differences across a row are preference effects and nothing else.

One row changes sign, and it is the contested one. Under a fixed real withdrawal the lead runs +21.91% at γ = 2, -2.08% at γ = 5 and -36.40% at γ = 10. Every other row in the table is positive at every risk aversion. So the reversal this paper reports is not a property of the means test on its own, and not of the withdrawal rule on its own: it needs a fixed real withdrawal *and* a household risk-averse enough for the loss of the floor to outweigh the equity premium. Between γ = 2 and γ = 5 the sign turns over.

That is a sharper statement of the paper's thesis than the one we set out to make, and a more conditional one. The interaction is not two-way but three-way, and the third term is the one a plan sponsor cannot observe. A default portfolio is chosen for a population, and this table says the right default in a means-tested system depends on where in that population the sponsor thinks the marginal member sits.

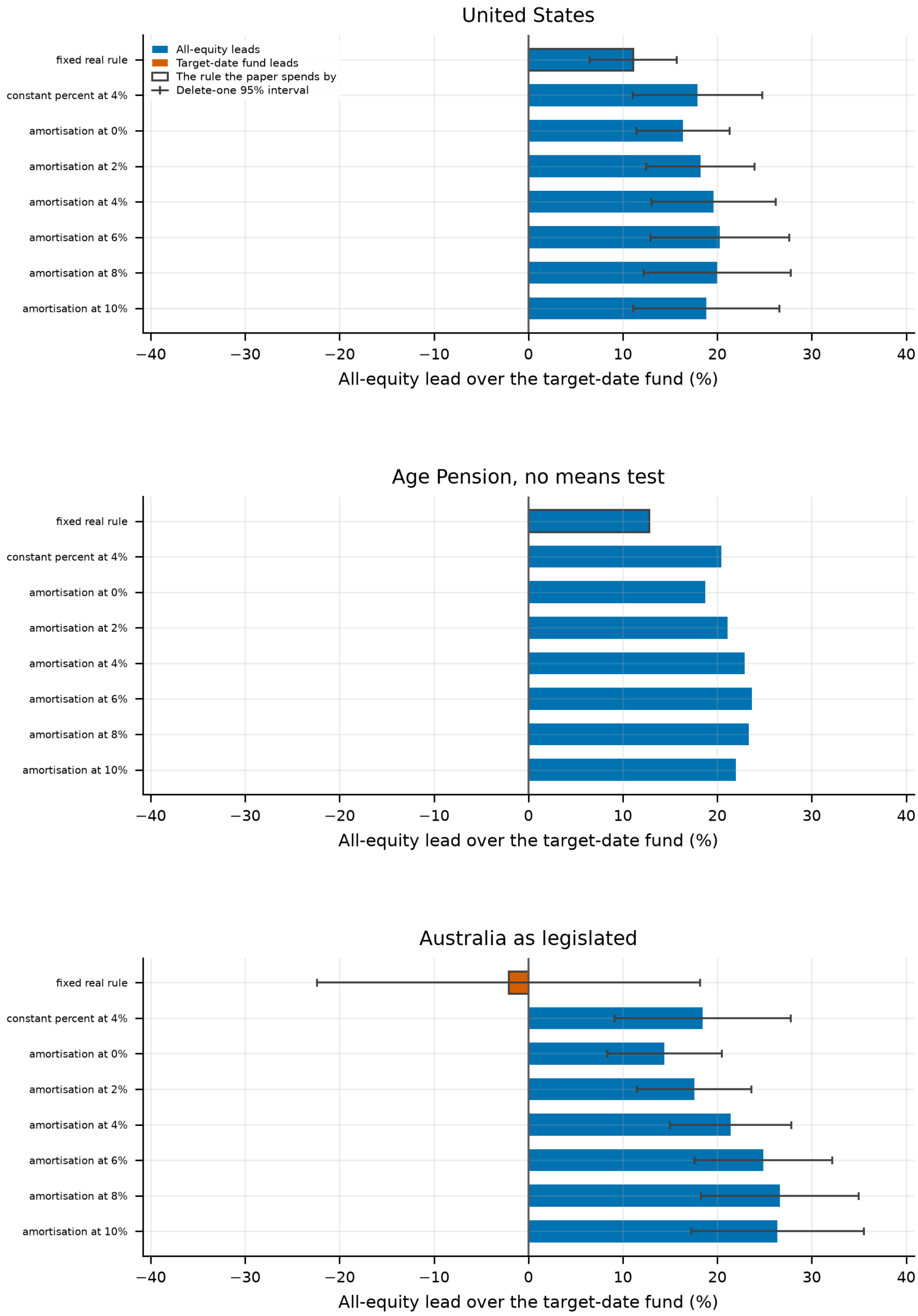

Figure 14.

The all-equity portfolio's lead over the target-date fund, by pension system and withdrawal rule. One panel per system on a shared scale; the zero line is the ranking, the outlined bar is the rule the rest of this paper spends by, and the whiskers are delete-one-country 95% intervals where they were computed.

11. What Would Change These Conclusions

The results above are the output of a calibrated simulation, and the honest question about any such thing is what would have to be different for it to come out the other way. Four things would.

11.1 The cross-section is sixteen countries

The panel is sixteen developed markets, and they are not sixteen independent draws: equity returns co-move, and the twentieth century happened to all of them at once. Dropping one country at a time moves the headline gap enough that the *direction* of every result here is far better established than its size. Across the 16 deletions the baseline gap runs +3.46% to +7.26%, and stays positive throughout. Nothing in this paper should be read as a point estimate, and the reversal claims are claims about signs.

11.2 One pension system, stylised

Australia's Age Pension is modelled as an assets test alone: a flat maximum rate, a free area, and a taper, re-assessed every retirement year against the drawn-down balance. The real system also applies an income test and a deeming rule, and pays the lower of the two results; it treats the family home as exempt, which moves a large share of Australian household wealth outside the test entirely; and it pays couples at a different rate on different thresholds. Each of those would change where a given household sits against the cut-off. None of them changes the shape of the budget line in Section 2, which is what the argument turns on — but a reader wanting the number for a particular Australian household will not find it here.

11.3 The household is one household

A single earner, a deterministic wage profile with a permanent shock, no unemployment, no career break, no divorce, no bequest motive beyond a fixed weight, and a home that is not modelled. Section 8 moves the balance across the whole width of the assets test, which addresses the objection that the results were demonstrated on a household the test could not bind; it does not turn one household into a distribution of them. The households that actually cluster near the cut-off differ from this one in more than their balance, and whether those differences reinforce or offset the effect measured here is not something this design can say.

11.4 No annuity, and no behaviour

The instrument that most directly supplies the floor this paper is about — a life annuity — is not in the choice set, and if it were it would dominate part of the grid. Nor is there any behavioural constraint: the household rebalances annually without hesitation, never panics, and follows its withdrawal rule for thirty years. The rule comparison in Section 9 should be read as what the rules deliver if followed, which is an upper bound on what they deliver.

11.5 What is settled, and what is not

Two things in this paper are robust to all of the above, because they are identities rather than estimates. Charging the Superannuation Guarantee to wages cannot change the balance at retirement, so it cannot change the retiree's problem — that is arithmetic, and the simulation reproduces it to every digit. And a withdrawal rule that spends a share of the balance converts a portfolio gain into consumption, where a fixed real rule converts it into assessable assets; the sign of that difference does not depend on the panel.

Everything else is an estimate. The fuller robustness apparatus — the bootstrap-design sweep, the out-of-sample split, the valuation conditioning, the fee and turnover ladders, the mortality calibration and the parameter sensitivities — is in the companion study, whose Sections 8 and 38 carry it. This paper inherits those results; it does not re-derive them.

12. Conclusion

The case for a fixed all-equity lifecycle portfolio is not one case but a family of them, and which member you are in is settled by two institutions and one preference rather than by the return process. Give an investor an earnings-related pension and a fixed real withdrawal and the all-equity portfolio leads the target-date fund by +11.09%. Change the pension to an asset-tested one and the lead becomes -2.08%: the de-risking glide path, the design the literature spends its length arguing against, wins. Change the withdrawal rule instead and it does not.

We would not have a reader stop at that sentence, because the panel does not support it. Sixteen countries put a standard error of 10.3 points on a gap of -2.08%, and the sign changes when eight of the sixteen are removed one at a time. The reversal is a point estimate this cross-section cannot separate from zero. What it can separate from zero, comfortably and in every cell, is the withdrawal rule's effect: within the Australian system alone the rule moves the all-equity lead by 29 percentage points, an order of magnitude larger than the gap in dispute.

A second check points the same way from a different direction. Re-scoring the identical simulated lifetimes at risk aversions of 2, 5 and 10, the contested cell runs +21.91% at $\gamma = 2$, -2.08% at $\gamma = 5$ and -36.40% at $\gamma = 10$ — it is not negative at the lowest one — while every other withdrawal rule in the menu stays positive throughout. The reversal therefore needs three things at once: an asset-tested pension, a fixed real withdrawal, and a household risk-averse enough to pay for the floor. Its absence needs only one of the three to fail.

That last sentence is the one we would put first. The reversal holds in 1 of the 8 withdrawal rules we run, and the rule it holds under is the fixed real withdrawal — which costs the same Australian household 52% of its certainty-equivalent consumption against the best rule in the same menu and runs out 12 percentage points more often. A retiree who follows Section 9's advice never meets the reversal. We report it because it is what the literature's standard assumption produces, not because we think a retiree should be in that cell.

The reason is worth separating from the obvious one. It is tempting to attribute the reversal to the means test's taper, which is steep enough to act as a wealth tax. It cannot be that: this household is past the cut-off before the test applies. It is the loss of the floor. Paying the same pension to everyone, with no test at all, leaves the all-equity portfolio ahead by +12.78% even though it pays this household less than the American schedule does. Withdrawing a pension at the margin and paying a smaller one are different interventions, and only the first reorders portfolios.

The second finding qualifies the first and should travel with it. A withdrawal rule that cannot deplete the portfolio supplies the floor the pension no longer does, and holding everything else fixed it moves the all-equity lead in the Australian system to +26.63% under amortisation at 8%, which returns the ordering the pension had reversed. So the reversal is not a fact about Australia. It is a fact about any retiree whose income in the bad states depends on the portfolio itself — which includes a saver under an asset test, and excludes one drawing on an annuity, a defined-benefit scheme, or a rule that divides by the years remaining.

The third finding is the one we did not expect and would now put first for a practitioner. Scaling the balance until the assets test genuinely binds — which no household in the sections above does, and which is the objection this design was built to answer — the portfolio the retiree wants stops being a fact about the pension at all. Spending a fixed real amount, they want no equity anywhere the test reaches. Spending a share of the current balance, they want 100% everywhere. The mechanism is not subtle: a rule that never spends what the portfolio earns turns a good return into assessable assets, and the pension is withdrawn against them while consumption stands still. Near a means test, the withdrawal rule does not modify the portfolio decision. It is the portfolio decision.

Three implications follow for default design. A plan sponsor choosing a glide path is implicitly choosing it against a pension schedule, and the same fund is not the right default in a means-tested system and an earnings-related one. The drawdown default matters as much as the portfolio default: in a system without a floor, supplying one through the withdrawal rule recovers what the pension no longer provides. And for the households a means test actually binds — which are not the wealthy ones the taper is usually discussed around — the two defaults cannot be set separately at all, because the rule decides whether an extra dollar of return reaches consumption or is taken back by the test.

We state these as implications of a calibrated model. It carries no behavioural constraints, prices no annuity, models one household rather than a distribution of them, and — as Section 11 sets out — rests on a sixteen-country panel whose leave-one-out range is wide enough that the direction of these results is far better established than their size. Two of them are exceptions, because they are identities rather than estimates: charging a compulsory contribution to wages cannot change the balance it produces, and a rule that spends a share of the balance converts a portfolio gain into consumption where a fixed real rule converts it into assessable assets. Neither depends on the panel.

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